ORIGINAL ARTICLE

Interpretable lightweight attention-guided deep learning framework for retinal optic disc and optic cup segmentation

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Abstract

Glaucoma is a critical eye condition that causes permanent blindness by damaging the Optic Nerve Head (ONH). Ophthalmologists diagnose glaucoma in patients by conducting morphological analysis of Optic Cup (OC) and Optic Disc (OD) regions in retinal fundus images. The implementation of lightweight and robust Artificial-Intelligence-enabled tools to deliver prompt glaucoma diagnostics is paramount in biomedical engineering. This article proposes CSP-SegNet, a novel Channel-Spatial-Pixel (CSP) attention-integrated lightweight encoder-decoder architecture for joint semantic segmentation of OC and OD in retinal fundus images. The crux of this work lies in the novel Channel-Spatial-Pixel (CSP) attention module which facilitates enhanced feature representation from different levels of abstraction with faster convergence. The efficacy of novel CSP attention is analyzed using Grad-CAM–based attention map evolution for explainable interpretation. CSP-SegNet is a novel depthwise separable convolutional neural network comprising approximately 1.54M trainable parameters with 13.3G FLOPS and hence it is highly lightweight compared to other methods. This paper has rigidly analyzed the robustness and generalization ability of CSP-SegNet in contrast to state-of-the-art segmentation networks, across the REFUGE and ORIGA datasets with cross-dataset evaluation on Drishti-GS. The proposed CSP-SegNet has obtained statistically significant results in outperforming many competing methods for joint semantic segmentation of OC and OD across Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) metrics. The quantitative and qualitative evaluation results justify the superiority of CSP-SegNet and effectiveness of CSP attention in terms of generalization ability, segmentation performance, robustness across distribution shift and compactness. The code is available at https://github.com/AIAnkitDas/CSP-SegNet/tree/main.

Keywords Attention · Lightweight encoder-decoder architecture · Deep neural network · Optic cup & optic disc segmentation · Explainable AI · Gradient-weighted class activation mapping (Grad-CAM)

1 Introduction

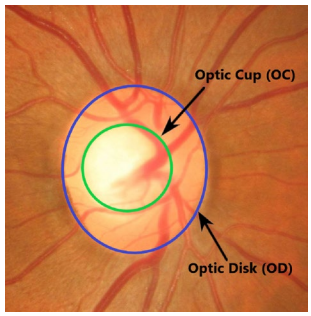
Human eyes suffer from multifarious diseases, and Glaucoma is a severe eye condition that leads to permanent blindness in patients. It is a neurodegenerative disease that is irreversible Tham et al. [1], and it affects almost 2.86% of the world's population above the age of forty years [2]. Glaucoma is an eye condition prevalent in middle-aged people and does not reflect any apparent symptoms during the early stages. However, some notable symptoms to identify Glaucoma include severe headache, redness and pain in the eyes and blurred vision. In most cases, patients only realize changes in their vision once it is too late. Glaucoma is, therefore, called the "silent thief of sight" as it surreptitiously causes vision loss [3].

Glaucoma is caused by the abnormal fluid balance that results in Intraocular Hypertension [4]. Human eyes constantly secrete aqueous humor, and excess aqueous humor is drained out through the drainage angle between the iris and the cornea. This continuous process builds the Intraocular Pressure (IOP) of the eye. Due to excessive secretion of aqueous humor the drainage path often gets blocked. The excess aqueous humor cannot drain out, resulting in the rise of IOP. This condition of high pressure within the eye is termed Intraocular Hypertension which consequently leads to the damage of the Optic Nerve Head (ONH) [4]. The ONH is an extension of our central nervous system, responsible for transmitting electrical impulses between the eyes and the brain. This abnormality is observed in glaucoma-affected patients where the ONH gets obliterated, and it ultimately leads to irreversible loss of eyesight.

Early diagnosis of glaucoma is quite intricate as it is asymptomatic during its infancy [5]. Hence, it becomes crucial for patients to undergo regular screening tests to avert adverse consequences. Thorough examination of intraocular eye pressure, drainage angle and the ONH is mandatory for glaucoma diagnosis. Some notable glaucoma diagnostic tests include visual field tests, Optical Coherence Tomography (OCT), measurement of corneal thickness and IOP [5–7]. However, these tests are relatively time-consuming, expensive and also subject to inter-observer variability. Besides, retinal fundus imaging has emerged as a fast, painless and cost-effective alternative that captures the retinal fundus. Therefore, image processing enabled paradigms can be deployed to deliver prompt glaucoma diagnostics. An example of a retinal fundus image is given in Fig. 1.

Glaucoma results in structural deformations of the retina, primarily the ONH, divided into Optic Cup (OC) and Optic Disc (OD) regions. Morphological analysis of the OC and OD is paramount for examining whether or not the optic nerve's head is damaged [8]. Therefore, the segmentation and localization of OC and OD is a pivotal task for any computer-aided diagnostic system. Since manual demarcation of OC and OD is a laborious and often inaccurate technique owing to limited availability of expert clinicians. Hence artificially intelligent diagnostic systems are quintessential for accurate and automatic detection of glaucoma. The segmentation of OC and OD plays a decisive role in extracting relevant biomarkers [9]. The Cup-to-Disc Ratio (CDR) is a fundamental biomarker widely used as the standard for glaucoma assessment. It is measured as the ratio of the vertical diameter of OC to the vertical diameter of OD [10, 11]. High CDR values are an indicator of glaucoma [12]. It is pertinent to note that segmenting OC is quite intricate compared to OD due to the presence of overlapping retinal vessels

Fig. 1 OC and OD of a retinal fundus image



and obscure boundary between the OC and OD [13]. Therefore, accurate semantic segmentation of OC and OD is of utmost significance.

This article addresses the problem of multi class semantic segmentation of OC and OD. The objective is to develop a novel, lightweight, and robust deep learning framework for the joint segmentation of optic cup and optic disc in retinal fundus images. This article proposes CSP-SegNet, a novel Channel-Spatial-Pixel (CSP) attention-integrated, lightweight encoder-decoder architecture capable of precisely segmenting the OC and OD in retinal fundus images. CSP-SegNet emerges as a computationally lightweight and efficient method in terms of optimized parameter-count and Floating-Point Operations per Second (FLOPS). The impactful contributions of this research work are outlined as follows:

1. The CSP-SegNet is an end-to-end trainable, lightweight, Channel-Spatial-Pixel (CSP) attention-integrated deep neural network for joint semantic segmentation of OC and OD in retinal fundus images, achieving state-of-the-art performance with only 1.54 M trainable parameters and 13.3 GFLOPS. CSP-SegNet's comprehensive overview is given in Sect. 3.2.
2. CSP-SegNet incorporates a novel attention mechanism called Channel-Spatial-Pixel (CSP) attention, which recalibrates important features from different levels of abstraction. The CSP attention extracts relevant semantic and contextual information from the channel, spatial, and pixel levels to capture the local and global level dependencies. It enhances feature representations and assists in faster convergence. CSP module is elaborately discussed in Sect. 3.2.3.
3. As discussed in Sect. 4.5, CSP-SegNet leverages the principles of depth-wise separable convolutions [14] instead of regular convolutions to substantially minimize the trainable parameters and consequently optimize the computational load.
4. The CSP-SegNet introduces Squeeze and Excitation Blocks (SEB) [15] followed by Inverted Residual Blocks (IRB) [16]. This ensures extraction of minute and fine-grained spatial information from the squeezed and excited features. SEB and IRB are described in Sects. 3.2.1 and 3.2.2, respectively.
5. Explainability is introduced by leveraging Gradient-weighted Class Activation Mapping (Grad-CAM) [17] to investigate the focus of our proposed CSP attention module. In Sect. 4.7 the efficacy of the CSP module is analyzed by visualizing the refinement of attended features at each level of skip connections within CSP-SegNet hierarchy. Algorithm 2 describes our Grad-CAM-based computation and attention visualization.
6. Rigorous experiments were conducted on the REFUGE [18] and ORIGA [19] databases with cross-dataset evaluation on Drishti-GS [20] to justify the robustness and generalizability of CSP-SegNet. CSP-SegNet outperforms competing state-of-the-art frameworks [21–28] in joint OC–OD segmentation, achieving statistically significant improvements on a majority. Furthermore, CSP-SegNet ensures more consistent and complete OC–OD contours. Quantitative and qualitative analysis is conducted in Sects. 4.4 and 4.6, respectively.

2 Literature review

This section outlines multifarious applications of Artificial Intelligence (AI) in the context of OC and OD segmentation. Contemporary research in the domain of medical imaging is dominated by Deep Learning (DL) [29]. This section briefly explores a plethora of DL-based methods contributing to the given agenda.

A. Sevastopolsky [30] proposed a pioneering research work that employed a simple U-Net-like architecture with fewer convolutional filters for OC and OD segmentation. Zilly et al. [31] proposed a novel entropy-based sampling technique along with a hierarchical Convolutional Neural Network (CNN) powered by an ensemble learning strategy. Wang et al. [32] leveraged the unsupervised domain adaptation principle to deploy a novel patch-based output space adversarial learning (pOSAL) algorithm for joint OC and OD segmentation. Researchers Liu et al. [33] presented a semi-supervised learning-based Conditional Generative Adversarial Network (CGAN) for joint segmentation of OC and OD. In addition, Bian et al. [34] implemented an attention-based cascaded network

guided by anatomical knowledge of OC and OD where they exploited CGAN to perform joint segmentation of OC and OD. Researchers Gu et al. [23] developed a generalized Context Encoder Network (CE-Net) to capture and preserve important high-level and low-level spatial features.

Vision Transformers (ViTs) [35] have emerged as the state-of-the-art in image recognition tasks since their inception [36]. Their application in the context of OC and OD segmentation has been proposed by researchers Yi et al. [37] in a Coarse-to-Fine Transformer Network (C2FTFNet) that incorporates U-Net and Circular Hough Transform (CHT), and a TransUNet3+. Furthermore, researchers Lu et al. [38] presented a Prior Knowledge-based Relation Transformer network (PKRT-Net) that introduces a Relation Transformer Fusion Module (RTFM) with prior knowledge of clinical vessels in deep learning for joint segmentation of OC and OD.

Let us investigate some lightweight DL methods for OC and OD segmentation. Firstly, Sun et al. [39] proposed a NAS (Neural Architecture Search) integrated nested U-shaped architecture that improves joint OC-OD segmentation while optimizing the model parameters to approximately 35M. Researchers Haider et al. [40] proposed Separable Linked Segmentation Residual Network (SLSR-Net) that comprises only 4.67M trainable parameters and achieves satisfactory segmentation performance. Liu et al. [41] presented image pyramid input with densely connected depthwise separable convolution network (DDSC-Net) for joint segmentation of OC and OD. Pachade et al. [42] developed NENet, a Nested EfficientNet architecture where EfficientNetB4 serves as an encoder with a nested decoder that is made up of Atrous Spatial Pyramid Pooling (ASPP) blocks and attention gates. Furthermore, Yuan et al. [43] proposed a multi-scale W-shaped CNN comprising only 7.32M parameters to perform joint segmentation of OC and OD by aggregating features from different receptive fields.

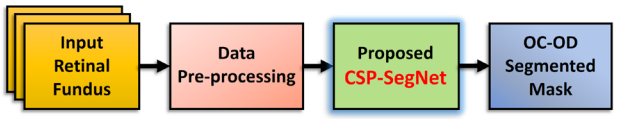
A few inspiring research articles pertinent to our methodology are delineated here. Researchers Hervella et al. [44] adopted a multi-task learning approach for simultaneous segmentation of OC and OD so as to retrieve the CDR and predict glaucoma. Tabassum et al. [45] proposed CDED-Net, a densely connected Cup Disc Encoder-Decoder Network for joint semantic segmentation of OC-OD. Wang et al. [46] presented EE-UNet, an Extended EfficientNet-based U-Net architecture for joint OC-OD segmentation. Here, the EfficientNet backbone is the feature extractor, which is extended to integrate Conditional Random Field as a Recurrent Neural Network (CRF-RNN). Researchers Hua et al. [47] proposed a domain generalization approach through DCAM-NET that substantially improves its generalization ability in terms of segmentation of OC and OD in retinal fundus images. The authors have introduced a novel Multi-Scale Attention (MSA) mechanism and Multi-Region Weight Fusion Convolution (MWFC) module that empowers the model to capture global-level semantics and adapt to unknown domain segmentation tasks. In addition Zhang et al. [48] proposed a transferable attention-based U-Net architecture for OC-OD segmentation. Finally, we have drawn immense inspiration from the research work conducted by Guo et al. [22], which proposed FAU-Net, a feature fusion and attention-integrated U-Net-like architecture for joint segmentation of OC and OD. The authors introduced a channel and spatial attention module (CSAM) within the decoder to obtain better segmentation results.

From this review we conclude that very few researchers have contributed to the development of lightweight, compact and robust OC-OD segmentation models. Our agenda to develop a novel, lightweight encoder-decoder framework for the joint segmentation of OC and OD is unique and efficient for mobile vision tasks. In addition, we introduce explainability by visualizing the Grad-CAM [17] generated attention heat-maps, to investigate the efficacy of our proposed mechanism.

3 Proposed methodology

This section defines the steps in developing our proposed framework for joint semantic segmentation of optic cup (OC) and optic disc (OD) in retinal fundus images. The Fig. 2 outlines our end-to-end workflow, from the retinal fundus imaging data to the final implementation of the deep learning framework that generates an OC-OD segmentation mask:

Fig. 2 End-to-end Workflow



Let us consider a 2D retinal fundus image R_i, represented as $\{x, R_i(x)|x \in \mathbb{R}^{(H\times W\times C)}\}$, where $R(x) : x \rightarrow \{0, 1, 2, \cdots, 255\}$ and H, W, C refers to the height, width and channel dimensions respectively. Here, R_i comprises three distinguishable regions – the Optic Cup (OC), the Optic Disc (OD), and the rest of the fundus. The primary aim of our proposed algorithm is to accurately localize the OC and OD regions from the rest of the fundus for a given R_i.

3.1 Data pre-processing

To interpret an image computationally, it first needs to be transformed into an array-like representation. Therefore pixel intensity values for each pixel in an image are mapped, and a corresponding matrix is obtained. To maintain consistency, all images need to be processed into a uniform shape. In our experiments, we have resized the retinal fundus image dimensions to 256 pixels in height and 256 pixels in width. This ensures uniformity in the shapes of the obtained features. It is imminent that the features be normalized to accelerate the training process. Rescaling was applied to convert the range of pixel intensity values to a smaller range of 0 to 1. This step improves the ability to generalize and brings efficiency to the learning process. Besides, the corresponding ground truths need to undergo categorical label encoding so that a label can be assigned to every pixel. In other words, black (representing OC), gray (representing OD), and white pixels (representing the rest of the fundus) are labeled and encoded as 0, 1 and 2 respectively. Our proposed network learns to map these patterns and predict the label corresponding to each pixel as black (0), gray (1), or white (2) to reconstruct the final segmentation mask.

3.2 Proposed CSP-SegNet

Traditionally, the utility of CNNs was only restricted to image recognition tasks. However, with the introduction of U-Net and Fully Convolutional Networks (FCN) in 2015 [49, 50], localization of regions of interest became a reality. In 2018, Attention U-Net [21] introduced the concept of integrating self-attention gating mechanisms. Since their inception, these traditional architectures have dominated numerous medical image segmentation tasks. Our proposed framework draws significant inspiration from conventional architectures but introduces novel advancements tailored for the joint segmentation of OC and OD.

In this article we propose CSP-SegNet, a novel Channel-Spatial-Pixel (CSP) attention-integrated, lightweight encoder-decoder architecture capable of accurately segmenting OC and OD in retinal fundus images. It is a fully connected depthwise separable convolutional neural network [14] that optimizes the computational load to a great extent. Each depthwise separable convolution layer is followed by batch normalization and rectified linear units (ReLU) [51] to mitigate the problem of vanishing and exploding gradient [52, 53]. CSP-SegNet comprises approximately 1.54M trainable parameters and 13.3 GFLOPS, underscoring its lightweight design and computationally efficient architecture. The proposed CSP-SegNet is a sequential architecture that constitutes an encoder module followed by a decoder module. The encoder follows the path of contraction to capture different scales of contextual information through multiple stages of down-sampling. On the contrary, the decoder serves as a symmetric expansion path that reconstructs the precise segmentation mask after multiple stages of up-sampling. The encoder module is directly connected to the decoder module at every stage, via a CSP attention-infused skip connection. These skip connections facilitate the preservation of enhanced contextual information lost during down-sampling. The CSP attention mechanism is fused with the skip connections to focus on the most significant diagnostic features across the channel as well as spatial and pixel levels of abstraction. Additionally, the network

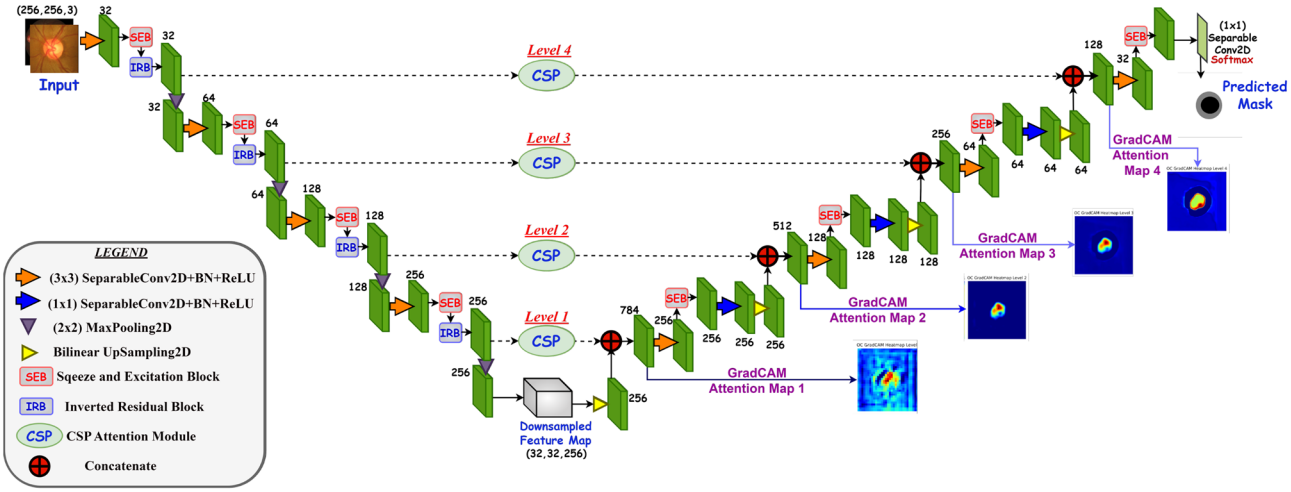
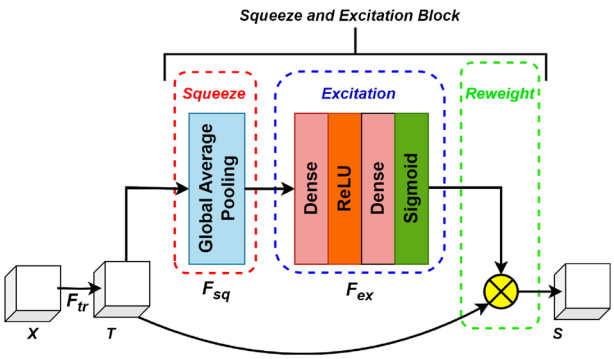


Fig. 3 Proposed CSP-SegNet architecture

Fig. 4 Squeeze and Excitation Block (SEB)



incorporates squeeze and excitation blocks [15] followed by inverted residual blocks [16] enabling it to encode robust OC-OD-aware features and adaptively recalibrate the representations. CSP-SegNet precisely reconstructs a segmentation mask to segregate the OC and OD regions for a given retinal fundus image. The comprehensive architecture of CSP-SegNet is demonstrated in Fig. 3.

3.2.1 Squeeze and Excitation Block (seb)

The SEB, derived from Squeeze-and-Excitation Networks [15], exploits an adaptive feature recalibration technique to enhance feature representations. It performs explicit modelling of inter-channel relationships to learn the globally significant information while suppressing the redundant ones. The schematic diagram of SEB is given in Fig. 4.

Consider an input feature representation $X \in \mathbb{R}^{(H_1 \times W_1 \times C_1)}$ that undergoes a given transformation to produce $T \in \mathbb{R}^{(H \times W \times C)}$. The transformation operation is represented by $F_{tr} : X \rightarrow T$. The transformed feature T initially undergoes a "squeeze" operation $F_{sq} : T \rightarrow C$ to aggregate (or squeeze) features across the $(H \times W)$ spatial dimensions using global average pooling. This enables the SEB to capture the global spatial information and generate a channel descriptor $C \in \mathbb{R}^{(1 \times 1 \times C)}$. Hence, the channel descriptor C embeds channel-wise feature responses from a global perspective.

The channel descriptor C then undergoes the "excitation" operation $F_{ex} : C \rightarrow C'$ on each channel to adaptively recalibrate the inter-channel dependencies through a self-gating mechanism. The resulting channel descriptor

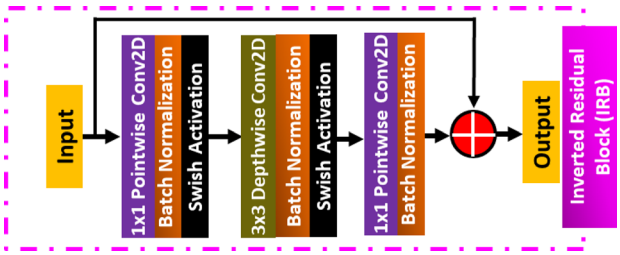


Fig. 5 Inverted Residual Block (IRB)

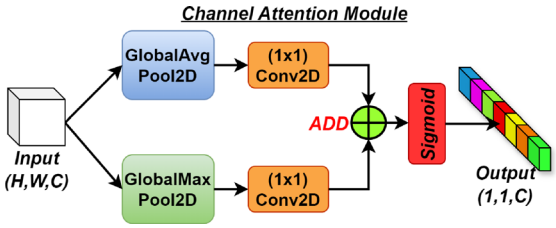


Fig. 6 Channel Attention (CA) module

$C' \in \mathbb{R}^{(1\times1\times C)}$ undergoes feature recalibration operation with the original feature T. Essentially, the feature maps T and C' are adaptively reweighted to produce $S \in \mathbb{R}^{(H\times W\times C)}$, which serves as the output of the SEB.

3.2.2 Inverted Residual Block (IRB)

The IRB is derived from the MobileNet architecture and the paper on inverted residuals and linear bottlenecks [16]. The IRB is a stacked topology of pointwise and depthwise convolution units powered by skip (residual) connections [54] to retain the high-level features. Within the CSP-SegNet encoder, the output of SEB is directly fed into the subsequent IRB. This ensures the extraction of minute and fine-grained features from the squeezed and excited feature. A schematic representation of IRB is given in Fig. 5. The primary attribute of IRB is the residual connection, which prevents loss of information and facilitates direct propagation of gradients in deep neural networks. This alleviates the problem of vanishing gradients and also helps achieve faster convergence. The term “inverted" refers to the fact that instead of expanding the dimensionality of feature maps (as in traditional convolutional blocks), the IRB first narrows it down, then applies lightweight depthwise convolutions, and finally expands the dimensionality back to the original size. Moreover, this inverted design accounts for considerable memory efficiency.

3.2.3 Channel-Spatial-Pixel (CSP) attention module

The proposed CSP is a novel, hybrid, self-supervised attention mechanism, capable of extracting powerful and robust OC-OD-aware features from delicate and obscure regions of the retinal fundus. The CSP attention module is adept at capturing relevant contextual information across the channel, spatial and pixel levels of abstraction. The CSP refines crucial features ranging from the high-level semantics to the low-level pixels by consolidating channel attention, spatial attention and pixel attention mechanisms [55, 56]. The proposed CSP attention module introduces a distinct architectural design and fusion strategy that differentiates it from existing approaches, and is relatively underexplored in the OC-OD segmentation problem. Unlike conventional attention modules that apply channel and spatial attention sequentially or independently, the CSP module integrates channel, spatial, and pixel-level attentions in a cooperative framework, enabling joint refinement of high-level semantic features and low-level pixel details. This coordinated fusion allows complementary information across different attention levels to be preserved and reinforced, rather than being suppressed by isolated gating operations. In the context of CSP-SegNet, the term “attention-integrated" refers to the incorporation of this hybrid CSP attention mechanism that facilitates better “understanding" of image-contents by highlighting features which map the underlying correlations among semantics [57]. The diagrams representing the channel attention, spatial attention and pixel attention are given in Figs. 6, 7 and 8 respectively.

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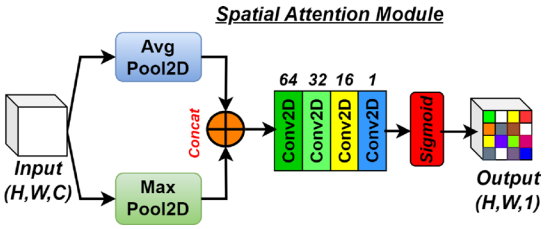


Fig. 7 Spatial Attention (SA) module

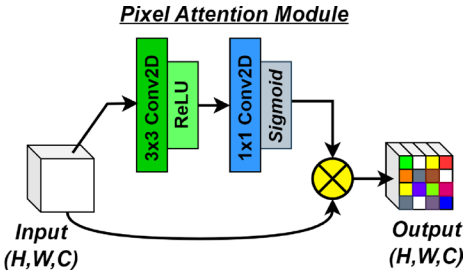
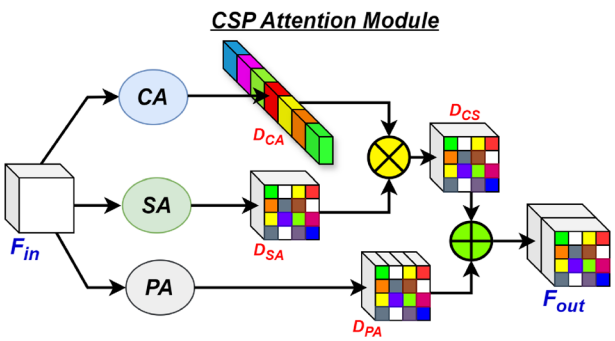


Fig. 8 Pixel Attention (PA) module

The Channel Attention (CA) module analyzes the inter-channel dependencies to compute an attention map that selectively focuses on the most pertinent features across each channel. This CA empowers our CSP attention module with a global receptive field. The Spatial Attention (SA) module analyzes the inter-spatial dependencies [58] to compute an attention map that selectively focuses on the most relevant local spatial features. The Pixel Attention (PA) module computes an attention map based on the contextual relevance of each pixel and thus operates at the lowest level of abstraction. The PA module is capable of precise boundary localization and capturing fine structural cues which is critical for mapping OC-OD contours. The proposed PA module employs a sequential combination of a 3×3 convolution followed by a 1×1 convolution powered by Sigmoid activation to model complementary pixel-level interactions. The 3×3 convolution captures localized spatial context and fine-grained structural patterns, which are critical for boundary delineation in dense prediction tasks. The subsequent point-wise convolution performs cross-channel interaction and adaptive re-weighting at each spatial location, enabling pixel-wise modulation of feature responses across channels. Unlike standard convolutional or residual blocks that aim to enhance representational capacity through feature transformation, the PA module explicitly learns an attention mask that selectively emphasizes informative pixels rather than generating new feature representations.

When combined with conventional channel (CA) and spatial attention (SA) mechanisms within the CSP framework, the pixel attention (PA) mechanism operates at a finer granularity. Channel attention (CA) highlights *what* semantic features are important, spatial attention (SA) identifies *where* these features are relevant at a coarse level, and pixel attention (PA) further refines *which specific pixels* within these regions should be emphasized. This hierarchical cooperation allows the network to preserve global semantic consistency while improving local boundary accuracy. Therefore, the output of the CSP attention module is an attention map that encompasses global-level semantic cues as well as low-level contexts. CSP module adopts an attention-based fusion technique by exploiting a channel-wise concatenation operation to preserve critical information derived from different levels of abstraction [57]. This CSP attention map highlights the distinguishable contextual information from regions of interest by suppressing the redundant ones. The schematic diagram of the CSP attention module is given in Fig. 9.

Fig. 9 Channel-Spatial-Pixel (CSP) attention module



Algorithm 1 CSP-SegNet Training Strategy

Given: Retinal fundus image $R_i \in \mathbb{R}^{H\times W\times C}$ and corresponding ground truth mask $M_i \in \{0,1,2\}^{H\times W}$; number of epochs E, batch size B, and learning rate α.
Ensure: CSP-SegNet model $\mathcal{M}$

1. Initialize CSP-SegNet with 1,541,594 trainable parameters.
2. Define loss function: $\mathcal{L} \in \{\text{Dice}, \text{Jaccard}, \text{Dice} + \text{Jaccard}\}$.
3. **AIM:** Minimize the loss function over training data:

$$\min_{\theta} \mathcal{L}(\theta)$$

4. Set optimizer: Adam($\alpha = 0.001$).
5. Set evaluation metrics: Precision, Recall, IoU, and DSC.
6. Define checkpoint callback to save the best model based on validation IoU.
7. **For** each epoch $e = 1$ to E **do**:
   (a) **For** each batch (x, y) in training set **do**:
      (i) Forward Pass: $\hat{y} \leftarrow \mathcal{M}(x)$.
      (ii) Compute loss: $\mathcal{L}(y, \hat{y})$.
      (iii) Backpropagate and update model weights.
   (b) Evaluate model on validation set.
   (c) Save model if validation IoU improves.
8. Generate Grad-CAM heatmaps to visualize attention on OC and OD regions at different levels.

Return: Best performing model $\mathcal{M}^*$

Consider a feature map $F_{in} \in \mathbb{R}^{(H\times W\times C)}$ is input to the CSP attention module. Here, the channel attention module generates a one-dimensional channel descriptor $D_{CA} \in \mathbb{R}^{(1\times 1\times C)}$ and the spatial attention module generates a two-dimensional spatial descriptor $D_{SA} \in \mathbb{R}^{(H\times W\times 1)}$. Additionally, the pixel attention module generates a three-dimensional pixel descriptor $D_{PA} \in \mathbb{R}^{(H\times W\times C)}$. Using the feature recalibration technique the descriptors D_{CA} and D_{SA} are projected onto a higher dimensional space. Element-wise multiplication between descriptors D_{CA} and D_{SA} outputs $D_{CS} \in \mathbb{R}^{(H\times W\times C)}$ as exhibited in Eq. 1:

$$D_{CS} = D_{CA} \otimes D_{SA} \tag{1}$$

Finally, the output of the CSP attention module $F_{out} \in \mathbb{R}^{(H\times W\times 2C)}$ is obtained after feature vectors D_{PA} and D_{CS} undergo channel-wise concatenation operation as exhibited in Eq. 2:

$$F_{out} = D_{CS} \oplus D_{PA} \tag{2}$$

The residual connections that originate from each level of encoder to connect the corresponding decoder is integrated with this CSP attention module. This facilitates efficient transmission of important high-level gradients leading to accelerated convergence.

3.3 Experimental setup and training strategy

The CSP-SegNet comprises 1,547,290 parameters of which 1,541,594 parameters are trainable, and the remaining 5696 parameters are non-trainable. *To ensure fair comparison and reliability in results, a consistent experimental environment was maintained across all experiments*. All experiments were conducted on a remote platform equipped with 29GB effective RAM and powered by an NVIDIA Tesla P100 GPU accelerator of 16GB

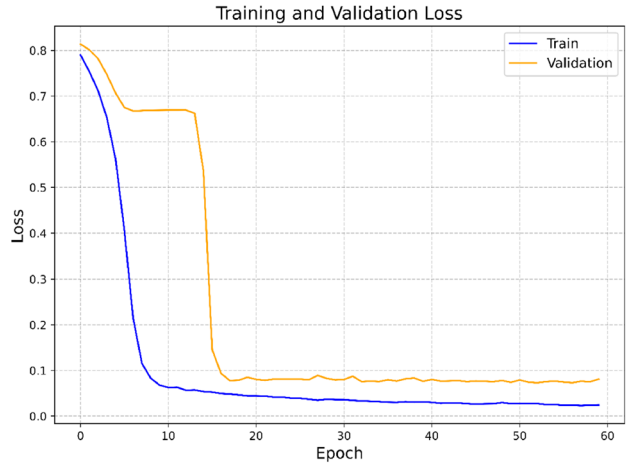


Fig. 10 CSP-SegNet on REFUGE with Dice Loss

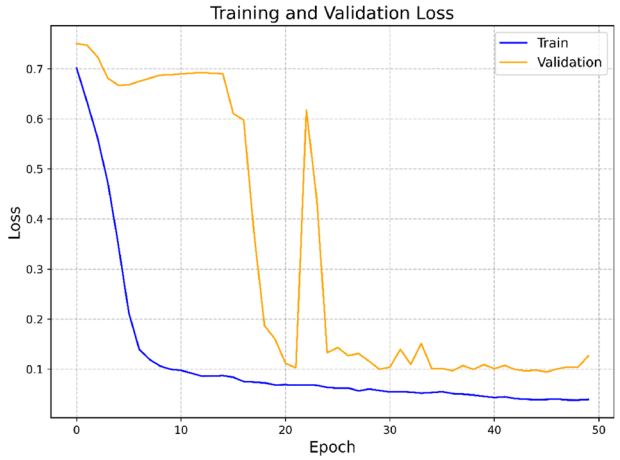


Fig. 11 CSP-SegNet on ORIGA with Dice Loss

high-bandwidth memory (HBM2). The algorithms were implemented in Python 3.10.12 with a TensorFlow 2.13.0 backend. After rigorous experiments with different hyper-parameters, we opted for compilation with Adam Optimizer [59] and a learning rate of 0.001. All models were trained for 60 epochs on REFUGE [18] and for 50 epochs on ORIGA [19] with a batch size of 16. Empirical process concluded that REFUGE [18], being a larger dataset, benefited from 60 epochs of training to fully converge, while ORIGA [19] is comparatively smaller showed signs of early convergence, making 50 epochs sufficient for optimal performance. The Algorithm 1 describes an end-to-end training pipeline adopted for CSP-SegNet.

3.4 Loss functions for CSP-SegNet

Since the joint semantic segmentation of OC and OD is a multi-class problem we have implemented multi-class region-aware loss functions to potentially improve the localization of OC and OD. For the optimization of CSP-SegNet, we have utilized the following loss functions:

3.4.1 Dice loss

Dice loss is a region-based loss function derived from the Dice Similarity Coefficient (DSC) [60, 61]. In the context of image segmentation, DSC measures the similarity between the predicted segmentation mask and the ground truth mask. The DSC for predicted mask p and the corresponding ground truth mask q is defined in Eq. 3:

$$DSC(p,q) = \frac{\sum_{i=1}^{N} p_i q_i + 1}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} q_i + 1} \tag{3}$$

Here, p_i and q_i represent the ith pixel of p and q respectively. Subsequently, dice loss is computed as the complement of the DSC shown in Eq. 4, and the aim is to minimize this loss function during training:

$$DiceLoss = 1 - \frac{\sum_{i=1}^{N} p_i q_i + 1}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} q_i + 1} \tag{4}$$

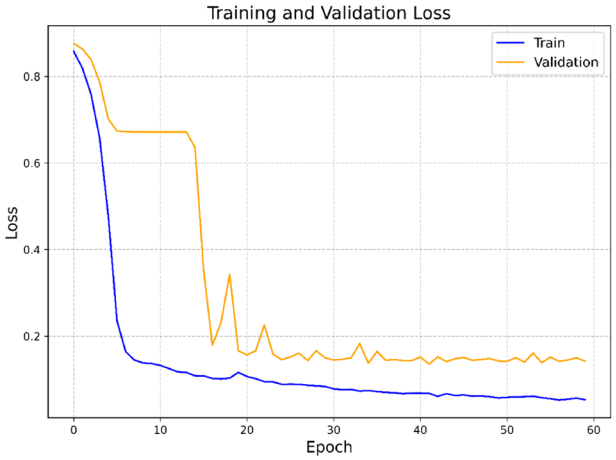


Fig. 12 CSP-SegNet on REFUGE with IoU Loss

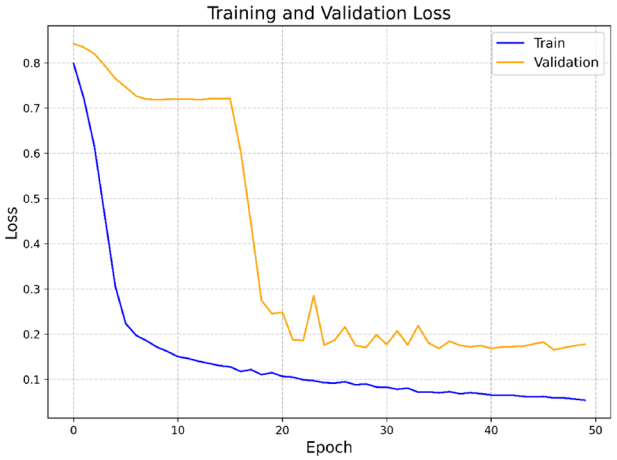


Fig. 13 CSP-SegNet on ORIGA with IoU Loss

The goal is to optimize the segmentation network with respect to the dice loss, encouraging it to produce segmentation masks that closely match the ground truth. The graphs in Figs. 10 and 11 illustrate CSP-SegNet's training and validation learning curves while optimizing dice loss, for REFUGE [18] and ORIGA [19] datasets respectively. These graphs prove that our proposed network aims to minimize the dice loss at each epoch.

3.4.2 Intersection over Union (IoU) loss

Jaccard loss, also known as Intersection over Union (IoU) loss, is derived from the Jaccard Index or IoU Score. The IoU score quantifies the similarity between predicted and ground truth segmentation masks as the ratio of intersection over union [62]. The Jaccard Index or IoU is computed as in Eq. 5:

$$IoU(p,q) = \frac{\sum_{i=1}^{N} p_i q_i}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} q_i - \sum_{i=1}^{N} p_i q_i} \tag{5}$$

Subsequently, the Jaccard loss or IoU loss is evaluated in Eq. 6 as the complement of the IoU:

$$IoULoss = 1 - \frac{\sum_{i=1}^{N} p_i q_i}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} q_i - \sum_{i=1}^{N} p_i q_i} \tag{6}$$

The goal during training is to minimize the Jaccard loss, encouraging the model to produce segmentation masks with a higher overlap with the ground truth. The graphs in Figs. 12 and 13 illustrates CSP-SegNet's training and validation learning curves while optimizing IoU loss, for REFUGE [18] and ORIGA [19] datasets respectively. These graphs prove that our proposed network aims to minimize the IoU loss at each epoch.

3.4.3 Combined loss

A combination of Dice Loss and Jaccard Loss aims to leverage the strengths of both loss functions. This potentially improves the model's ability to capture different aspects of segmentation performance [63]. The combined

 https://doi.org/10.1007/s00521-026-12036-9

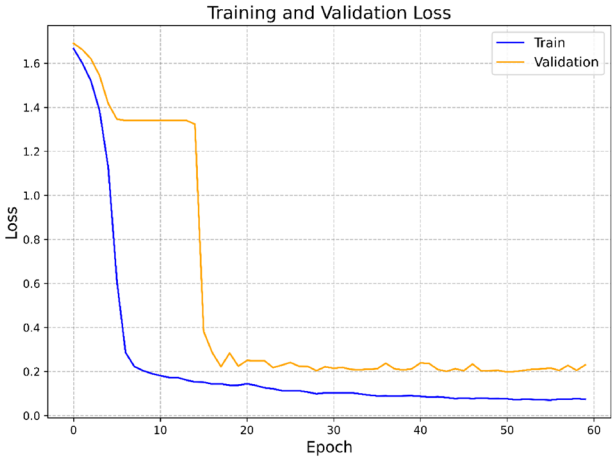


Fig. 14 CSP-SegNet on REFUGE with Combined Loss

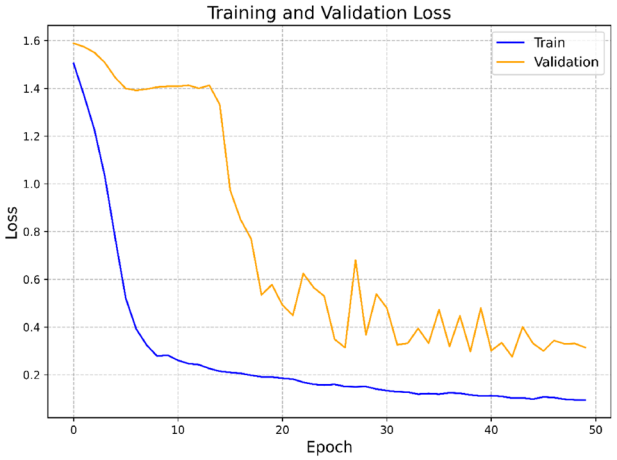


Fig. 15 CSP-SegNet on ORIGA with Combined Loss

loss function is expressed as the sum of the Dice loss and Jaccard loss. The combined loss can be defined as in Eq. 7:

$$CombinedLoss = DiceLoss + IoULoss \tag{7}$$

The graphs in Figs. 14 and 15 illustrates CSP-SegNet's training and validation learning curves while optimizing this combined loss, for REFUGE [18] and ORIGA [19] datasets respectively. These graphs prove that our proposed network aims to minimize the combined loss at each epoch.

The graphs illustrated in Figs. 10 to 15 demonstrates how the our proposed CSP-SegNet minimizes the loss after every epoch. These graphs explain the dynamics and convergence behaviour of our model and how different loss functions influence training stability and generalization. Comparative analysis of these graphs highlight that the Dice loss has proven to be impactful in faster convergence and reduced overfitting, across both datasets, as evident from the training and validation curves. Here we may infer that Dice loss yeilds effective and stable learning performance, although IoU loss and the combined loss also demonstrate reasonably competitive results.

4 Experimental results and discussion

When it comes to medical imaging applications, there is very little room for error. That being said, the need for robust, artificially intelligent computer imaging frameworks is paramount. This section evaluates the performance of our proposed OC-OD segmentation method on three publicly available datasets [18–20]. Furthermore, the performance of the proposed CSP-SegNet is compared with existing state-of-the-art OC-OD segmentation networks [21–28] across these datasets. *All experiments were conducted under a consistent experimental environment to guarantee fair and reliable performance evaluation*. This section presents a quantitative, qualitative, and statistical evaluation of the robustness, generalizability, reliability, and computational efficiency of CSP-SegNet's segmentation performance in comparison with competing methods.

4.1 Dataset description

The experiments were conducted on three advanced retinal fundus datasets. Our primary focus was on the REFUGE [18] and ORIGA [19] datasets followed by cross-dataset validation on Drishti-GS [20], to demonstrate the robustness and generlizability of our model.

The REFUGE dataset [18] is a contribution of the "Retinal Fundus Glaucoma Challenge", held in conjunction with MICCAI, 2018 and obtained from the EURECOM AML 2021: Challenge 2 [64]. It is a comprehensive collection of 1200 annotated retinal fundus images captured using 'Canon CR-2' and 'Zeiss Visucam 500' cameras. The ORIGA dataset [19] is a retinal fundus database developed and annotated by experts of the Singapore Eye Research Institute, under the Singapore Malay Eye Study (SiMES). The REFUGE [18] and ORIGA [19] dataset comprises a total of 1200 and 650 annotated retinal images, respectively. Additionally, the Drishti-GS [20] dataset comprising 50 training and 51 test expert annotated retinal fundus images were collected at the Aravind Eye Hospital, Madurai. Furthermore, to assess the generalization capability, we have conducted cross-dataset validation, using the ORIGA [19] dataset for training and validation, and the Drishti-GS [20] dataset for testing purpose. This cross-dataset evaluation helps in studying robustness across distribution shift. The distribution of datasets into training, validation and test sets adopted in our experiments is given in Table 1:

4.2 Evaluation metrics

In these experiments, four different metrics were computed to quantitatively evaluate the segmentation performance of each model. These four metrics - Precision, Recall, Dice Similarity Coefficient (DSC) and IoU Score (or Jaccard Index) are expressed in terms of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) which are formalized in Eqs. 8, 9, 10 and 11 respectively:

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

$$DSC = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{10}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{11}$$

Table 1 Dataset Split

Dataset	Training Set	Validation Set	Test Set
REFUGE [18]	540	60	600
ORIGA [19]	450	50	150
Drishti-GS [20]	450 (ORIGA)	50 (ORIGA)	51 (Drishti-GS)

Table 2 Performance of CSP-SegNet on REFUGE [18] and ORIGA [19] test sets with Different Loss Functions. Best results are in bold

	REFUGE [18]			ORIGA [19]		
Metrics	Dice Loss	IoU Loss	Combined Loss	Dice Loss	IoU Loss	Combined Loss
DSC_{OC}	0.8853	0.8806	**0.8911**	**0.8759**	0.8748	0.8724
DSC_{OD}	0.8964	0.8928	**0.8986**	0.8515	**0.8575**	0.8377
IoU_{OC}	0.7942	0.7866	**0.8036**	**0.7792**	0.7775	0.7737
IoU_{OD}	0.8122	0.8064	**0.8158**	0.7414	**0.7506**	0.7207

Table 3 Ablation Study of Proposed CSP-SegNet Configurations for OC–OD Segmentation on the REFUGE [18] Dataset. Configuration C1 represents the backbone and Configuration C8 represents the end-to-end CSP-SegNet architecture (Results in bold)

Configuration	Core Components			Metrics			
	CSP	SEB	IRB	DSC_{OC}	DSC_{OD}	IoU_{OC}	IoU_{OD}
C1	✗	✗	✗	0.8350	0.8533	0.7168	0.7442
C2	✓	✗	✗	0.8661	0.8614	0.7638	0.7565
C3	✗	✓	✗	0.8492	0.8609	0.7378	0.7558
C4	✗	✗	✓	0.8370	0.8299	0.7197	0.7092
C5	✓	✓	✗	0.8601	0.8785	0.7545	0.7833
C6	✗	✓	✓	0.8445	0.8824	0.7309	0.7895
C7	✓	✗	✓	0.8557	0.8674	0.7477	0.7659
C8	✓	✓	✓	**0.8853**	**0.8964**	**0.7942**	**0.8122**

Table 4 Ablation Study of Proposed CSP-SegNet Configurations for OC–OD Segmentation on the ORIGA [19] Dataset. Configuration C1 represents the backbone and Configuration C8 represents the end-to-end CSP-SegNet architecture (Results in bold)

Configuration	Core Components			Metrics			
	CSP	SEB	IRB	DSC_{OC}	DSC_{OD}	IoU_{OC}	IoU_{OD}
C1	✗	✗	✗	0.7514	0.7527	0.6017	0.6035
C2	✓	✗	✗	0.8308	0.7829	0.7106	0.6434
C3	✗	✓	✗	0.8532	0.8108	0.7439	0.6817
C4	✗	✗	✓	0.7750	0.7842	0.6326	0.6450
C5	✓	✓	✗	0.8533	0.8292	0.7442	0.7082
C6	✗	✓	✓	0.8550	0.8205	0.7468	0.6957
C7	✓	✗	✓	0.8363	0.8264	0.7186	0.7040
C8	✓	✓	✓	**0.8759**	**0.8515**	**0.7792**	**0.7414**

4.3 Ablation study

4.3.1 Impact of different loss functions

The proposed CSP-SegNet optimized the dice loss, IoU loss and combined loss functions as shown in subsection 3.4. The results exhibited in Table 2 reports the impact of different loss functions on the segmentation performance of CSP-SegNet across the REFUGE [18] and ORIGA [19] test sets.

From Table 2 we conclude that CSP-SegNet achieves the best region-aware segmentation scores on REFUGE [18] dataset when optimized with combined loss. Interestingly, on the ORIGA [19] dataset, it is observed that CSP-SegNet achieves the highest Dice and IoU scores for optic cup (OC) segmentation when optimized with Dice loss, while it attains superior Dice and IoU scores for optic disc (OD) segmentation when optimized with IoU loss.

4.3.2 Ablation study on CSP-SegNet network configurations

Experiments were conducted to examine the efficacy of the cohesive modules associated with our proposed CSP-SegNet. In Tables 3 and 4 we analyzed the contribution of these components in the improvement of segmentation performance on the REFUGE [18] and ORIGA [19] datasets, respectively. The analysis is based on different network configurations of CSP-SegNet derived from all possible combinations of its associated modules i.e., CSP attention module, Squeeze and Excitation Block (SEB) and Inverted Residual Block (IRB). This establishes eight network configurations, starting with C1 as the backbone eliminating all the core blocks, to C8 as the complete CSP-SegNet architecture. Configurations C2 to C7 represents combinations of CSP, SEB and IRB in different capacities.

The ablative results in Tables 3 and 4 for both REFUGE [18] and ORIGA [19], consistently demonstrate that the progressive integration of CSP attention, Squeeze-and-Excitation block, and Inverted Residual Block, leads to substantial improvements in OC-OD segmentation performance. Reasonably, the backbone-only configuration (C1) yields the lowest Dice and IoU scores for both datasets. C2 introduces the CSP attention to the backbone, which registers considerable improvement in segmentation performance across both datasets. CSP improves OC Dice by 3.11% on REFUGE [18] and 7.94% on ORIGA [19]. Among single-module variants, CSP attention (C2) and SEB (C3) provide larger gains than IRB alone (C4), highlighting the dominant role of attention-driven channel and spatial recalibration in OC-OD contour delineation. The C5 configuration introduces both the CSP as well as the SEB to the backbone, which registers an improvement in OC Dice by 2.51% on REFUGE [18] and by 10.19% on ORIGA [19]. The C6 configuration introduces the SEB with the IRB to the backbone. C6 significantly improves OD Dice by 2.91% on REFUGE [18] and by 6.78% on ORIGA [19]. This shows that SEB and IRB together act as a robust OC-OD feature extraction unit. Notable performance improvement is observed in C8 which corresponds to the original CSP-SegNet architecture that includes all three core components. C8 improves OC Dice by a considerable margin of 5.03% on REFUGE [18] and 12.45% on ORIGA [19], over backbone (C1).

These consistent gains across two distinct datasets validate that the proposed CSP attention, together with SEB and IRB, synergistically enhances both region-level discrimination and boundary localization, resulting in a robust and generalizable OC–OD segmentation framework. It is evident from the results exhibited in Tables 3 and 4 that all three modules are collectively responsible for achieving the best segmentation performance on both REFUGE [18] as well as ORIGA [19] datasets. Furthermore, these results manifest a significant degree of cohesiveness among the constituent modules of CSP-SegNet. Thus, the efficacy of the CSP attention module and the combination of the squeeze and excitation block (SEB) along with an inverted residual block (IRB) is justified.

4.3.3 Impact of variation in attention mechanisms

The impact of replacing the CSP attention module with existing sophisticated attention mechanisms within the CSP-SegNet hierarchy is analyzed in Table 5. In this context, experiments were conducted on the REFUGE [18] dataset, with the sophisticated Convolutional Block Attention Module (CBAM) [55], Squeeze and Excitation Block (SE Block) [15], and Pixel Attention [56].

Table 5 Impact of Variation in CSP-SegNet's Attention Mechanism on OC–OD Segmentation Performance evaluated on the REFUGE [18] dataset. The CSP attention is replaced with CBAM [55], SE Block [15] and Pixel Attention [56] within the CSP-SegNet hierarchy. Best results are in bold

Attention Mechanism	DSC_{OC}	DSC_{OD}	IoU_{OC}	IoU_{OD}
With CBAM [55]	0.8384	0.8591	0.7218	0.7530
With SE Block [15]	0.8690	0.8722	0.7684	0.7733
With Pixel Attention [56]	0.8624	0.8685	0.7581	0.7676
CSP-SegNet (With CSP)	**0.8853**	**0.8964**	**0.7942**	**0.8122**

The results exhibited in Table 5 provide clear quantitative evidence of the superiority of CSP attention module over the sophisticated CBAM, SE Block and Pixel attention mechanisms. CSP-SegNet consistently achieves the highest Dice and IoU scores for both OC and OD. It is observed that after CSP-SegNet, the SE Block [15] and Pixel attention [56] mechanisms register competitive OC-OD segmentation performance, while the CBAM [55] falls short by a considerable margin. Specifically, the SE Block [15] achieves the second best results with OC Dice of 0.8690 i.e., behind CSP-SegNet by a margin of 1.63%. Falling marginally short of SE Block [15] is the Pixel attention achieving an OC Dice of 0.8624 i.e., 2.29% behind CSP-SegNet. The CSP module achieves a balanced enhancement of channel, spatial, and pixel-wise feature representations, enabling more precise region consistency and OC-OD boundary delineation. These results confirm that the proposed CSP attention is more effective than existing attention mechanisms when integrated into the same CSP-SegNet hierarchy.

4.3.4 Ablation study on CSP attention configurations

To investigate the efficacy of individual attention mechanisms within the cohesive CSP attention module, an ablation study was conducted by selectively enabling and disabling the channel, spatial, and pixel attention branches. As shown in Table 6, a total of eight CSP configurations were evaluated, ranging from the attention-free baseline (CSP-0) to the complete CSP module integrating all three attention mechanisms. Intermediate variants include single-attention (CSP-C, CSP-S, CSP-P) and dual-attention (CSP-CS, CSP-SP, CSP-CP) configurations, allowing a systematic assessment of how each attention type and their combinations influence OC–OD segmentation performance on the REFUGE [18] dataset.

The results in Table 6 clearly indicate that each attention component contributes to progressive improvement in OC-OD segmentation performance. Reasonably the attention-less baseline CSP-0 exhibits the lowest Dice and IoU scores. In partial configurations, it is observed that the pixel attention has proven to be highly beneficial in improving segmentation performance, as is evident in CSP-P, CSP-SP and CSP-CP. The full CSP module, which integrates channel, spatial, and pixel attention, consistently outperforms all partial configurations, achieving the highest Dice and IoU scores for both OC and OD. The CSP-CP emerges as the second best configuration as it records an OC Dice of 0.8779 which is short of CSP's 0.8853 by a margin of only 0.74%. Besides, just enabling the pixel attention in configuration CSP-P, yields an OC Dice of 0.8708 which shows a gain of 2.63% over the attention-free baseline CSP-0. These results confirm that the joint modeling and recalibration of important features across channel, spatial, and pixel levels of abstraction is essential for precise delineation of OC-OD contours, validating the design of the proposed CSP attention mechanism.

4.4 Quantitative evaluation

This section quantitatively analyzes the segmentation performance of CSP-SegNet against other sophisticated segmentation networks namely – Attention U-Net [21], FAU-Net [22], CE-Net [23], CS-Net [24], Trans-UNet

Table 6 Ablation Study on Varying CSP Attention Configurations (channel, spatial, and pixel) for OC–OD Segmentation on the REFUGE [18] Dataset. Original CSP results are in bold

Configuration	Attention Mechanisms			Metrics			
	Channel	Spatial	Pixel	DSC_{OC}	DSC_{OD}	IoU_{OC}	IoU_{OD}
CSP-0	✗	✗	✗	0.8445	0.8824	0.7309	0.7895
CSP-C	✓	✗	✗	0.8559	0.8717	0.7481	0.7726
CSP-S	✗	✓	✗	0.8639	0.8856	0.7604	0.7947
CSP-P	✗	✗	✓	0.8708	0.8881	0.7712	0.7987
CSP-CS	✓	✓	✗	0.8663	0.8757	0.7641	0.7788
CSP-SP	✗	✓	✓	0.8606	0.8770	0.7554	0.7810
CSP-CP	✓	✗	✓	0.8779	0.8909	0.7824	0.8033
CSP	✓	✓	✓	**0.8853**	**0.8964**	**0.7942**	**0.8122**

[25], Swin-UNet [26], MISSFormer [27], and FCBFormer [28]. All quantitative metrics are reported as the mean and standard deviation, reflecting both the average performance and the stability of each model across experimental runs. In addition, statistical significance testing was conducted using paired t-test, and the resulting p-values are reported to assess whether the performance differences between CSP-SegNet and competing methods are statistically meaningful i.e., with $p < 0.05$ indicating statistically significant improvements. This statistical significance testing ensures reliability of the claimed performance improvements observed for the proposed CSP-SegNet method. It is pertinent to note that *CSP-SegNet is treated as the reference model for all statistical comparisons*. Statistical significance tests (p-values) are computed by comparing each baseline method against CSP-SegNet; therefore, no p-value is reported for the proposed method.

The comparative results for the joint segmentation of OC and OD on the REFUGE [18] dataset are presented in Table 7 to analyze the segmentation performance of CSP-SegNet in contrast to other state-of-the-art segmentation networks.

It is evident from the comparative results in Table 7 that our proposed CSP-SegNet outperforms the given state-of-the-art segmentation models across different evaluation metrics on the REFUGE [18] dataset. CSP-SegNet has achieved an average DSC of 0.8746 and 0.8866, besides an average IoU score of 0.7772 and 0.7964 for the joint segmentation of OC and OD, respectively. In contrast, CS-Net [24] has emerged as the second best performer registering an average DSC of 0.8546 and 0.8734, and an average IoU score of 0.7461 and 0.7754 for segmenting OC and OD, respectively. Trans-UNet [25] falls marginally short of CS-Net [24], which yielded a mean DSC of 0.8463 and 0.8628, and a mean IoU score of 0.7343 and 0.7590 for segmenting OC and OD, respectively. CE-Net [23] and FCBFormer [28] have relatively underperformed attaining inferior DSC and IoU scores. FAU-Net [22] exhibits considerable divergence in precision and recall (3.16% approx.), indicating a strong over-segmentation tendency that introduces excessive false positives. This behavior is undesirable for any consistent diagnostic framework and is also weakly observed in Trans-UNet [25] and Swin-UNet [26] but is effectively mitigated in CSP-SegNet. Therefore, the results prove that our proposed method has significantly overpowered the compared state-of-the-art methods. Furthermore, the paired t-test results indicate that the performance gains achieved by CSP-SegNet over several competing methods are statistically significant ($p < 0.05$), including Attention U-Net [21], CE-Net [23], CS-Net [24], Swin-UNet [26], and FCBFormer [28]. Although the differences with FAU-Net [22], Trans-UNet [25], and MISSFormer [27] are not statistically meaningful under the paired t-test ($p > 0.05$), CSP-SegNet consistently attains higher mean DSC and IoU scores across both OC and OD classes. This indicates that CSP-SegNet delivers superior segmentation accuracy in both quantitative and qualitative terms, while the lack of statistical significance in these cases can be attributed to variance across test samples and the limited size of the evaluation set rather than inferior model performance.

Table 7 Quantitative Analysis of OC–OD Segmentation Performance of CSP-SegNet against state-of-the-art methods on the REFUGE [18] test set. All models are trained and validated on the dedicated REFUGE splits and evaluated on its test set. The Best, 2nd Best, and Worst Scores are highlighted in Underlined, Bold, and Bold-italic, respectively. Statistically significant results ($p<0.05$) are *italicized*

Network	Precision	Recall	DSC_{OC}	DSC_{OD}	IoU_{OC}	IoU_{OD}	p-value
Attention U-Net [21]	0.8982 ± 0.0344	0.9037 ± 0.0297	0.8362 ± 0.0060	0.8539 ± 0.0150	0.7186 ± 0.0088	0.7453 ± 0.0229	*0.014*
FAU-Net [22]	***0.8655*** ± ***0.0205***	0.8971 ± 0.0272	0.8280 ± 0.0230	0.7996 ± 0.0184	0.7069 ± 0.0336	0.6663 ± 0.0256	0.063
CE-Net [23]	0.8835 ± 0.0477	***0.8534*** ± ***0.0604***	***0.8091*** ± ***0.0296***	***0.7723*** ± ***0.0640***	***0.6801*** ± ***0.0419***	***0.6321*** ± ***0.0874***	*0.040*
CS-Net [24]	0.9001 ± 0.0144	**0.9208** ± **0.0089**	**0.8546** ± **0.0062**	**0.8734** ± **0.0117**	**0.7461** ± **0.0094**	**0.7754** ± **0.0183**	*0.009*
Trans-UNet [25]	0.8956 ± 0.0248	0.9149 ± 0.0081	0.8463 ± 0.0290	0.8628 ± 0.0204	0.7343 ± 0.0430	0.7590 ± 0.0312	0.243
Swin-UNet [26]	0.8879 ± 0.0017	0.9156 ± 0.0072	0.8413 ± 0.0103	0.8611 ± 0.0034	0.7261 ± 0.0152	0.7560 ± 0.0052	*0.038*
MISSFormer [27]	0.8878 ± 0.0220	0.8894 ± 0.0469	0.8397 ± 0.0344	0.8244 ± 0.0703	0.7246 ± 0.0502	0.7052 ± 0.0987	0.171
FCBFormer [28]	**0.9003** ± **0.0176**	0.8817 ± 0.0201	0.8266 ± 0.0192	0.8420 ± 0.0221	0.7047 ± 0.0281	0.7275 ± 0.0326	*0.023*
CSP-SegNet (ours)	0.9175 ± 0.0066	0.9245 ± 0.0060	0.8746 ± 0.0020	0.8866 ± 0.0056	0.7772 ± 0.0032	0.7964 ± 0.0090	*

*Statistical significance tests (p-values) are computed by comparing each baseline method against the reference method, CSP-SegNet; therefore, no p-value is reported for the proposed method

Table 8 Quantitative Analysis of OC–OD Segmentation Performance of CSP-SegNet against state-of-the-art methods on the ORIGA [19] test set. All models are trained and validated on the dedicated ORIGA splits and evaluated on its test set. The Best, 2nd Best, and Worst Scores are highlighted in Underline, Bold, and Bold-italic, respectively. Statistically significant results ($p<0.05$) are *italicized*

Network	Precision	Recall	DSC_{OC}	DSC_{OD}	IoU_{OC}	IoU_{OD}	p-value
Attention U-Net [21]	**0.8963 ± 0.0093**	**0.8888 ± 0.0127**	**0.8566 ± 0.0145**	**0.8237 ± 0.0077**	**0.7494 ± 0.0221**	**0.7003 ± 0.0111**	0.124
FAU-Net [22]	0.8654 ± 0.0198	0.8397 ± 0.0119	0.8013 ± 0.0172	0.7659 ± 0.0233	0.6688 ± 0.0241	0.6211 ± 0.0308	*0.045*
CE-Net [23]	0.9005 ± 0.0093	0.8189 ± 0.0359	0.8106 ± 0.0198	0.7631 ± 0.0507	0.6818 ± 0.0280	0.6187 ± 0.0648	*0.038*
CS-Net [24]	0.8958 ± 0.0053	0.8777 ± 0.0079	0.8491 ± 0.0060	0.8152 ± 0.0160	0.7379 ± 0.0090	0.6883 ± 0.0227	0.282
Trans-UNet [25]	0.8665 ± 0.0188	0.8470 ± 0.0507	0.8101 ± 0.0367	0.7618 ± 0.0564	0.6822 ± 0.0522	0.6175 ± 0.0729	0.113
Swin-UNet [26]	0.8660 ± 0.0041	0.8409 ± 0.0127	0.7846 ± 0.0170	0.7678 ± 0.0105	0.6458 ± 0.0228	0.6232 ± 0.0138	*0.006*
MISSFormer [27]	0.8751 ± 0.0167	0.8501 ± 0.0231	0.7952 ± 0.0386	0.7882 ± 0.0256	0.6612 ± 0.0538	0.6510 ± 0.0351	*0.046*
FCBFormer [28]	***0.8234 ± 0.0050***	***0.7967 ± 0.0177***	***0.7033 ± 0.0129***	***0.7210 ± 0.0123***	***0.5425 ± 0.0153***	***0.5639 ± 0.0151***	*0.001*
CSP-SegNet (ours)	0.8991 ± 0.0039	0.8967 ± 0.0038	0.8637 ± 0.0125	0.8383 ± 0.0118	0.7602 ± 0.0194	0.7217 ± 0.0176	∗

∗ Statistical significance tests (p-values) are computed by comparing each baseline method against the reference method, CSP-SegNet; therefore, no p-value is reported for the proposed method

Table 8 presents the comparative results for the joint segmentation of OC and OD on the ORIGA [19] dataset. These results aim to assess the segmentation efficacy of CSP-SegNet in comparison to other state-of-the-art segmentation networks.

The results exhibited in Table 8 suggest the superiority of our proposed CSP-SegNet over existing state-of-the-art segmentation networks across all evaluation metrics on the ORIGA [19] dataset. In contrast to the REFUGE [18] dataset, the ORIGA [19] dataset is more challenging. It is observed that CSP-SegNet has achieved a mean DSC of 0.8637 and 0.8383, besides mean IoU score of 0.7602 and 0.7217 for the segmentation of OC and OD, respectively. We observe that Attention U-Net[21] emerges as the second best performer securing a mean DSC of 0.8566 and 0.8237, along with mean IoU score of 0.7494 and 0.7003 for segmentation of OC and OD, respectively. Closely following Attention U-Net [21] is the CS-Net [24], achieving a mean DSC of 0.8491 and 0.8152, besides mean IoU score of 0.7379 and 0.6883 for the segmentation of OC and OD, respectively. The Trans-UNet [25], CE-Net [23] and FAU-Net [22] methods show closely clustered results exhibiting consistently lower DSC and IoU values with higher variability, indicating that these models produce nearly similar and inferior segmentation outcomes. In contrast, it is observed that the FCBFormer [28] has failed miserably recording the lowest DSC and IoU scores, indicating that it struggles to capture the domain-specific OC-OD structural details of the ORIGA [19] images. FAU-Net [22] and particularly CE-Net [23] exhibits notable divergence between precision and recall, with comparatively high precision but substantially lower recall, indicating missed OC and OD regions due to a low false-positive rate but a high number of false negatives [65]. Such behavior is undesirable for reliable clinical deployment as it risks excluding true pathological structures. A similar but weaker trend is also observed in Trans-UNet [25] and Swin-UNet [26], whereas CSP-SegNet maintains a well-balanced precision–recall trade-off, ensuring more complete and stable delineation of both OC and OD. Therefore, it is evident from these results that our proposed framework shows considerable improvement in segmentation performance. Furthermore, the paired t-test results indicate that CSP-SegNet achieves statistically significant performance improvements ($p < 0.05$) over FAU-Net [22], CE-Net [23], Swin-UNet [26], MISSFormer [27], and FCBFormer [28]. In contrast, the performance differences with Attention U-Net [21], CS-Net [24], and Trans-UNet [25] are not statistically significant ($p > 0.05$), despite CSP-SegNet attaining higher mean DSC and IoU scores for both OC and OD. This suggests that while CSP-SegNet consistently provides superior segmentation accuracy on ORIGA [19] in both quantitative and qualitative aspects, the lack of statistical significance in few cases is likely due to inter-sample variability and dataset-specific characteristics rather than a limitation of the proposed architecture.

Table 9 reports the cross-dataset evaluation results for joint OC–OD segmentation on the Drishti-GS [20] dataset, where all models undergo training and validation on ORIGA [19] and testing on Drishti-GS [20] test set to explicitly assess generalizability under distribution shift. This cross-dataset experimental setting evaluates the

Table 9 Quantitative Analysis of OC–OD Segmentation Performance of CSP-SegNet against state-of-the-art methods on the Drishti-GS [20] test set. All models are trained and validated on the dedicated ORIGA [19] splits and evaluated on the Drishti-GS [20] test set. The Best, 2nd Best, and Worst Scores are highlighted in Underline, Bold, and Bold-italic, respectively. Statistically significant results ($p<0.05$) are *italicized*

Network	Precision	Recall	DSC_{OC}	DSC_{OD}	IoU_{OC}	IoU_{OD}	p-value
Attention U-Net [21]	0.8014 ± 0.0255	0.7367 ± 0.0773	**0.7524 ± 0.0459**	0.5295 ± 0.1094	**0.6044 ± 0.0579**	0.3650 ± 0.0994	0.308
FAU-Net [22]	0.7977 ± 0.0154	0.5711 ± 0.0005	0.5380 ± 0.0352	0.3703 ± 0.0343	0.3680 ± 0.0220	0.2272 ± 0.0184	*0.042*
CE-Net [23]	**0.8186 ± 0.0525**	***0.5325 ± 0.0132***	***0.5185 ± 0.0228***	0.2924 ± 0.0286	***0.3499 ± 0.0203***	0.1714 ± 0.0195	*0.010*
CS-Net [24]	0.7553 ± 0.0344	0.6173 ± 0.0261	0.6862 ± 0.0077	0.1451 ± 0.0725	0.5224 ± 0.0089	0.0793 ± 0.0416	*0.010*
Trans-UNet [25]	***0.7194 ± 0.0193***	0.6432 ± 0.0243	0.7354 ± 0.0267	***0.1340 ± 0.0387***	0.5821 ± 0.0332	***0.0722 ± 0.0222***	*0.049*
Swin-UNet [26]	0.8112 ± 0.0119	0.7371 ± 0.0368	0.7515 ± 0.0368	0.5457 ± 0.0720	0.6028 ± 0.0469	0.3776 ± 0.0702	0.201
MISSFormer [27]	0.8303 ± 0.0192	**0.7895 ± 0.0295**	0.7257 ± 0.0653	**0.6726 ± 0.0224**	0.5722 ± 0.0792	**0.5070 ± 0.0257**	0.225
FCBFormer [28]	0.8133 ± 0.0140	0.7810 ± 0.0776	0.7408 ± 0.0942	0.6492 ± 0.0700	0.5940 ± 0.1144	0.4832 ± 0.0772	0.084
CSP-SegNet (ours)	0.8026 ± 0.0226	0.8241 ± 0.0045	0.7842 ± 0.0101	0.6881 ± 0.0254	0.6450 ± 0.0137	0.5248 ± 0.0298	∗

∗ Statistical significance tests (p-values) are computed by comparing each baseline method against the reference method, CSP-SegNet; therefore, no p-value is reported for the proposed method

robustness of CSP-SegNet and competing methods when exposed to unseen data acquired under different imaging protocols, thereby assessing whether a model generalizes learned representations across datasets.

The cross-dataset evaluation results obtained from Table 9 clearly demonstrates the superior generalization capability of the proposed CSP-SegNet under a pronounced distribution shift from ORIGA [19] to Drishti-GS [20]. CSP-SegNet achieves the highest mean DSC of 0.7842 and 0.6881 along with the highest mean IoU of 0.6450 and 0.5248 for OC and OD segmentation, respectively. Among the competing methods, MISSFormer [27] emerges as the second best performer in terms of OD segmentation, obtaining a mean DSC of 0.7257 and 0.6726 and a mean IoU of 0.5722 and 0.5070 for OC and OD, respectively. Besides, Attention U-Net [21] emerges as the second best performer in terms of OC segmentation but with notably lower OD accuracy and higher variance, registering a mean DSC of 0.7524 and 0.5295 and a mean IoU of 0.6044 and 0.3650 for OC and OD, respectively. FCBFormer [28] follows closely, attaining a mean DSC of 0.7408 and 0.6492 and a mean IoU of 0.5940 and 0.4832 for OC and OD, respectively. In contrast, FAU-Net [22], CE-Net [23], CS-Net [24], and Trans-UNet [25] exhibit severe performance degradation, particularly for OD segmentation, with markedly low DSC and IoU values, indicating limited robustness to domain shift and difficulty in capturing cross-dataset OC-OD characteristics. CE-Net [23] miserably fails to segment OC pixels, whereas Trans-UNet [25] miserably fails to segment OD pixels. Notably, majority of competing methods display pronounced divergence between precision and recall, which may lead to over-segmentation of erroneous regions or missed segmentation of OC-OD pixels. This behavior is undesirable for any clinical system, where completeness of structure delineation is critical. In contrast, CSP-SegNet maintains a well-balanced precision–recall trade-off and exhibits consistently lower variance, reflecting its ability to preserve both accuracy and stability across distribution shifts. Furthermore, paired t-test results show that CSP-SegNet achieves statistically meaningful ($p < 0.05$) improvements over FAU-Net [22], CE-Net [23], CS-Net [24], and Trans-UNet [25], confirming that the observed performance gains under cross-dataset evaluation are not by chance. Although the differences with Attention U-Net [21], Swin-UNet [26], MISSFormer [27], and FCBFormer [28] are not statistically significant ($p > 0.05$), CSP-SegNet consistently attains higher mean DSC and IoU for both OC and OD, indicating superior robustness and generalizability under distribution shift rather than overfitting to the source domain.

4.5 Analysis of computational load

The introduction of depthwise separable convolutions instead of regular convolutions into our proposed CSP-SegNet has substantially reduced the number of trainable parameters thereby optimizing the computational cost [61]. When CSP-SegNet was compiled with regular convolutions the resulting number of trainable parameters

Table 10 Computational Load Analysis of CSP-SegNet against state-of-the-art Methods in Terms of Parameters (in millions) and FLOPS (in Giga-FLOPS). Best and worst results are in bold and italics, respectively

Network	Parameters (in Millions)	FLOPS (in GFLOPS)
Attention U-Net [21]	7.94M	25.0G
FAU-Net [22]	15.32M	*300.3G*
CE-Net [23]	*40.93M*	17.0G
CS-Net [24]	8.40M	25.1G
Trans-UNet [25]	12.47M	19.6G
Swin-UNet [26]	3.62M	**10.8G**
MISSFormer [27]	18.91M	12.58G
FCBFormer [28]	4.95M	77.63G
Proposed CSP-SegNet	**1.54M**	13.3G

were approximately 4.13M. The same network when the regular convolutions were replaced with depthwise separable convolutions, resulted in 1.54M trainable parameters. A reduction in almost 2.59M parameters significantly accounts for our proposed framework's compactness. This is because depth-wise separable convolutions apply only a single filter per input channel compared to traditional convolutions. Hence, depthwise separable convolution accounts for our CSP-SegNet's lightweight feature. A comparative study based on the total number of trainable parameters (in millions) and floating-point operations per second (FLOPS expressed in giga-FLOPS) is given in Table 10.

It is evident from Table 10 that our proposed CSP-SegNet is the lightest of all in terms of total number of trainable parameters. CSP-SegNet constitutes the lowest number of parameters i.e., 1.54M only, with relatively optimal FLOPS count of 13.3 GFLOPS only. Hence, CSP-SegNet is much more optimized to be efficiently deployed into mobile devices. On the contrary, CE-Net [23] is the heaviest model exhibiting the highest parameter-count of 40.93M, whereas FAU-Net [22] records the highest computational complexity with 300.3 GFLOPS. Overall, CSP-SegNet uniquely combines model compactness and segmentation accuracy, achieving the lowest model size with relatively optimal computational cost, while still delivering the best DSC and IoU scores for joint OC and OD segmentation. This balance between computational load and dominant quantitative performance makes CSP-SegNet particularly well suited for real-time and resource-constrained clinical deployment without sacrificing diagnostic reliability.

4.6 Qualitative evaluation

This section qualitatively analyzes the segmentation performance of CSP-SegNet over compared state-of-the-art segmentation networks – Attention U-Net [21], FAU-Net [22], CE-Net [23], CS-Net [24], Trans-UNet [25], Swin-UNet [26], MISSFormer [27], and FCBFormer [28]. A visual comparative study on the predicted OC-OD segmented masks generated by each deep learning framework is outlined for critical test data-points in REFUGE [18], ORIGA [19] and Drishti-GS [20] datasets.

Figure 16 illustrates the segmentation results produced by CSP-SegNet and competing state-of-the-art segmentation networks on few challenging REFUGE [18] test data points. These qualitative results are coherent with the results exhibited in Table 7.

Figure 17 illustrates the segmentation results produced by CSP-SegNet and competing state-of-the-art segmentation networks on few challenging ORIGA [19] test data points. These qualitative results are coherent with the results exhibited in Table 8.

Figure 18 illustrates the segmentation results produced by CSP-SegNet and competing state-of-the-art segmentation networks on few challenging Drishti-GS [20] test data points. These qualitative results are coherent with the results exhibited in Table 9.

The visual depictions in Figs. 16, 17 and 18 illustrate the segmentation performance of different state-of-the-art networks against our proposed method, on some relatively challenging test cases. A thorough qualitative

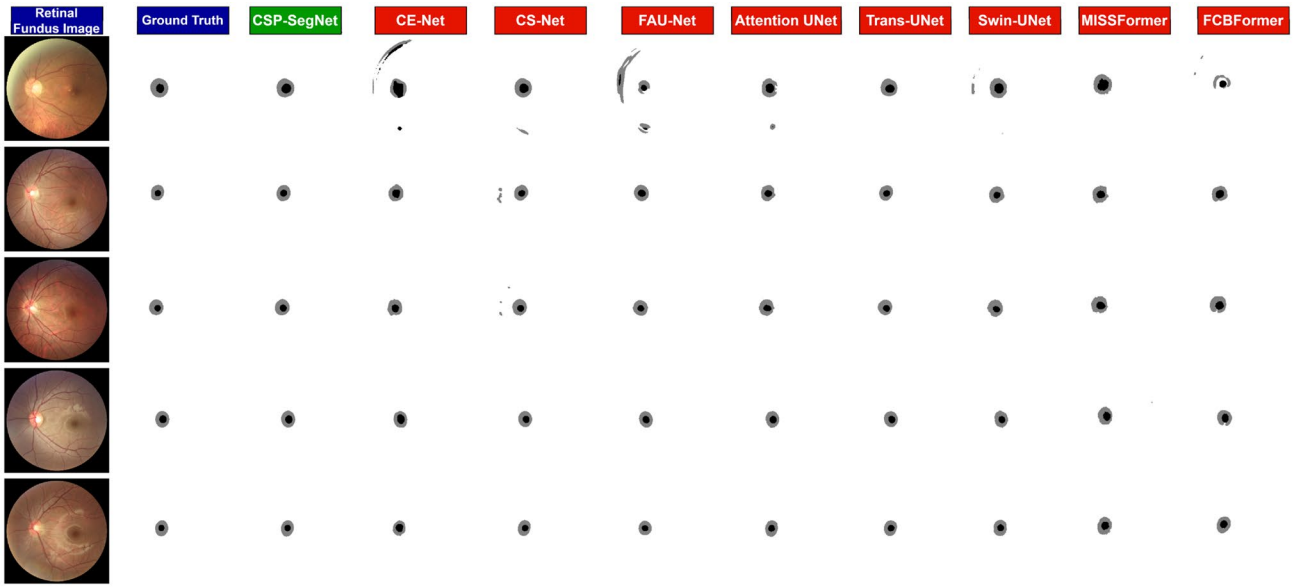


Fig. 16 Qualitative analysis of OC–OD segmentation performance of CSP-SegNet against state-of-the-art methods on the REFUGE [18] dataset

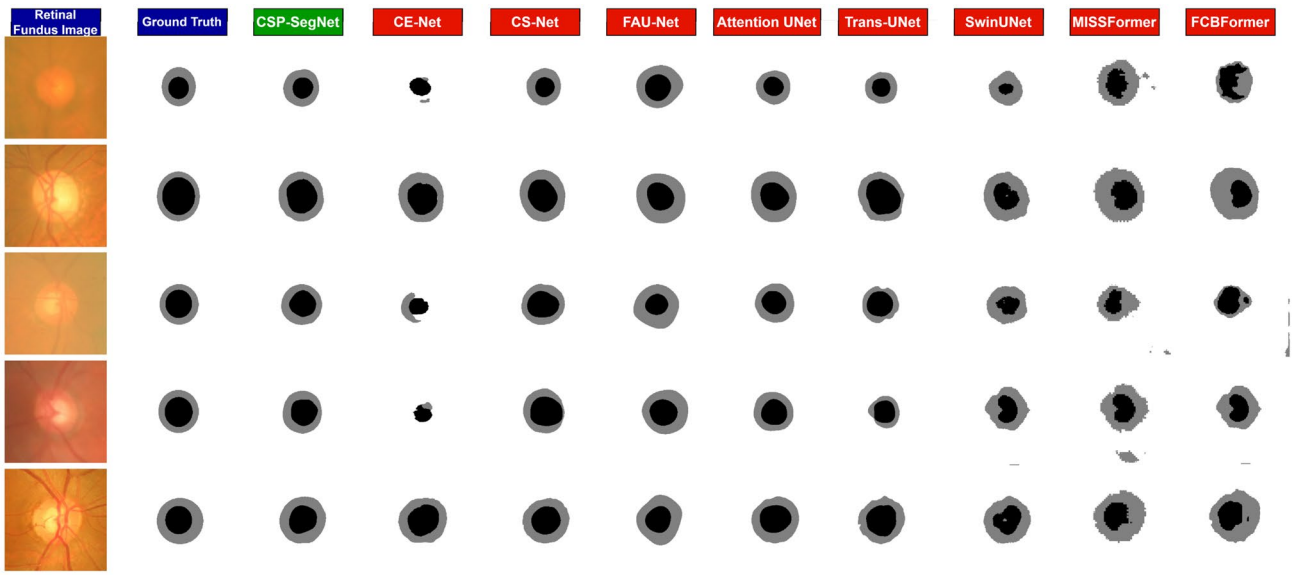


Fig. 17 Qualitative analysis of OC–OD segmentation performance of CSP-SegNet against state-of-the-art methods on the ORIGA [19] dataset.

evaluation suggests that most of the existing state-of-the-art encoder-decoder architectures struggle to distinguish complicated, scattered and diffused OC-OD boundaries. Our proposed CSP-SegNet has shown remarkable improvement in OC-OD segmentation performance on REFUGE [18], ORIGA [19] and Drishti-GS [20] datasets. CSP-SegNet correctly predicts the target OC-OD regions that exhibit very nominal difference with the ground truth. Furthermore, our network accumulates very minimal false negatives compared to other networks and thus considerably reduces the false positive predictions. It is visually perceived that CSP-SegNet preserves the OC-OD contours to ensure a smoother, complete and prominent boundary between OC and OD throughout its predictions. In Fig. 16, it is observed that CS-Net [24], Attention-UNet [21] and Trans-UNet [25] exhibit relatively accurate

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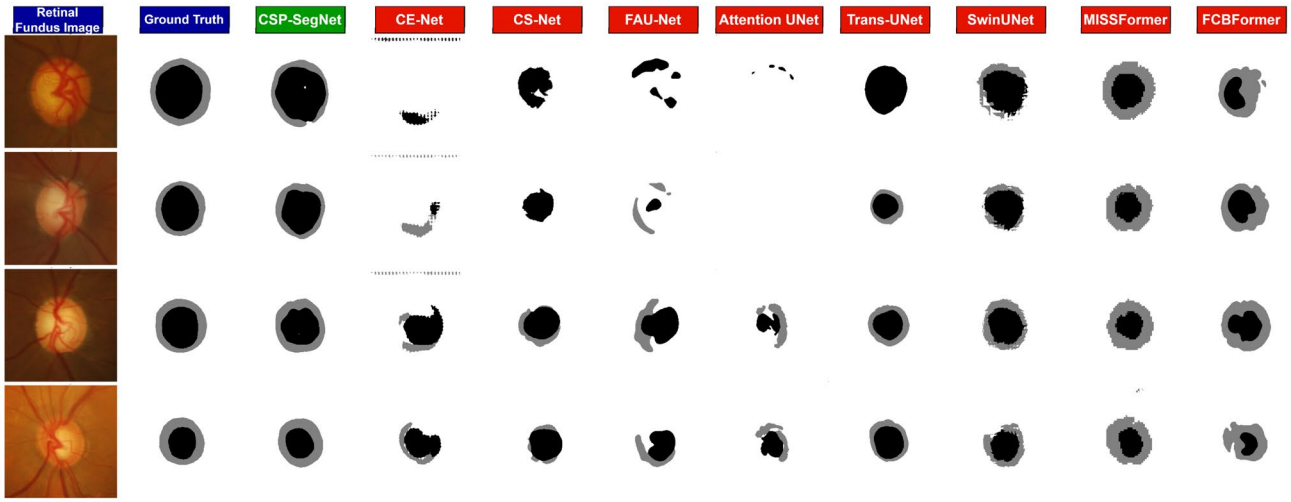


Fig. 18 Qualitative analysis of the cross-dataset generalizability of CSP-SegNet and state-of-the-art methods for OC–OD segmentation on the Drishti-GS [20] dataset

OC–OD delineations that are visually close to the ground truth, although their segmentations remain consistently less precise and less stable than those produced by CSP-SegNet. Qualitative analysis of Fig. 16 also suggests that FAU-Net [22] erroneously segments outlier pixels as OC–OD regions, which aligns with its strong precision–recall divergence and indicates unstable segmentation with missed regions and spurious predictions. In Fig. 17, it is observed that Attention U-Net [21] and CS-Net [24] generate visually reasonable OC–OD delineations that are broadly consistent with the ground truth, although their predictions remain less accurate and less stable than those produced by CSP-SegNet. Qualitative inspection on Fig. 17 further reveals that CE-Net [23] struggles to consistently capture the true OC–OD extents, often leading to incomplete or distorted boundaries, while FAU-Net [22] produces irregular and fragmented OC–OD contours. Evidently MISSFormer [27] exhibits erroneous region predictions owing to the precision-recall divergence. In contrast, FCBFormer [28] yields the poorest segmentation maps with severe mislocalization and loss of anatomical structure, highlighting its inability to adapt to the domain-specific characteristics. Qualitative inspection on Fig. 18 suggests that MISSFormer [27] produces visually strong OC–OD segmentations under distribution shift, although its boundaries are comparatively less sharp. CSP-SegNet maintains decent and anatomically consistent localization of both OC and OD across datasets, indicating robust generalization across distribution shift. In contrast, CS-Net [24] and Trans-UNet [25] predominantly captures the optic cup while failing to reliably delineate the optic disc, leading to severely imbalanced segmentations. CE-Net [23] exhibits erroneous and fragmented predictions, which aligns with its strong precision–recall divergence observed in the quantitative analysis. Although FCBFormer [28] yields satisfactory visual performance after MISSFormer [27], its OC–OD contours remain irregular and anatomically inconsistent, reflecting limited robustness under cross-dataset distribution shift. Overall, CSP-SegNet precisely segments intricate test cases showing significant differences in OC and OD sizes. It establishes less noisy segmentation masks enriched with smoother contours to minimize variations with ground truth. It is observed that CSP-SegNet consistently outperforms existing deep learning frameworks for the joint segmentation of OC and OD. CSP-SegNet can distinguish between sharp details to efficiently segment OC and OD in retinal fundus images and is proven to be robust to distribution shift.

Figure 19 presents the qualitative ablation study of CSP-SegNet on the ORIGA [19] dataset, illustrating how the progressive integration of the Channel-Spatial-Pixel (CSP) attention module, Squeeze-and-Excitation Block (SEB), and Inverted Residual Block (IRB) influences OC–OD segmentation behavior. The Fig. 19 visualizes the segmentation outputs of all eight configurations (C1–C8), ranging from the backbone network to the complete

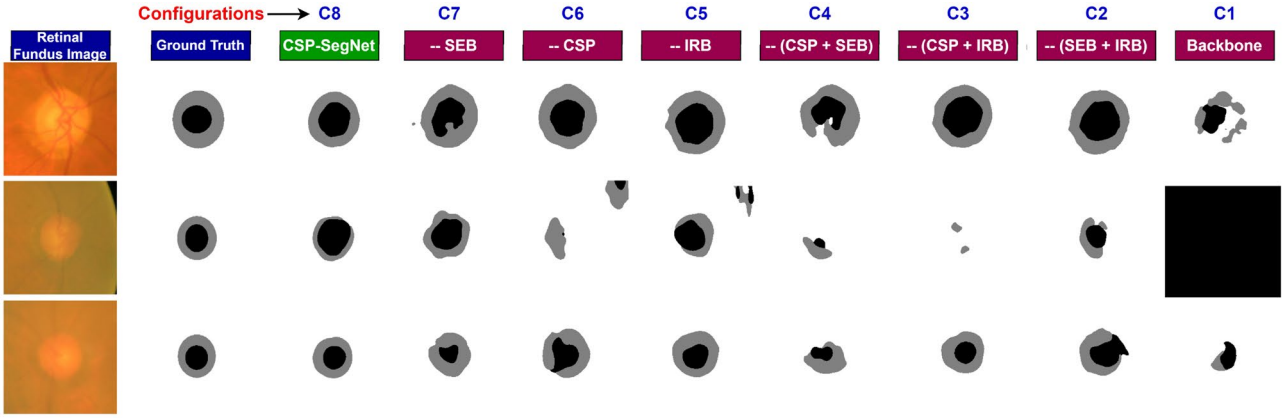


Fig. 19 Qualitative ablation study of CSP-SegNet for OC–OD segmentation performance on the ORIGA [19] dataset, illustrating the impact of individual architectural components

Algorithm 2 Grad-CAM-based attention visualization with CSP-SegNet

Given: Best performing model $\mathcal{M}^*$, input fundus image R_i, target class $c \in \{\text{OC}, \text{OD}\}$
Ensure: Grad-CAM heatmaps $\mathcal{H}_c^{(l)}$ for each skip level l

1. Forward pass R_i through $\mathcal{M}$ to obtain prediction $\hat{M}_i$
2. **for** each skip connection level $l = 1$ to 4 **do**:
   (a) Identify the corresponding convolutional feature map $A^{(l)}$
   (b) Compute gradients $\frac{\partial \mathcal{M}_c}{\partial A^{(l)}}$
   (c) Global average pool the gradients to obtain weights: $\alpha_k^{(l)} = \frac{1}{Z} \sum_{i,j} \frac{\partial \mathcal{M}_c}{\partial A_{ijk}^{(l)}}$
   (d) Compute weighted combination: $L_{\text{Grad-CAM}}^{(l)} = \text{ReLU}\left(\sum_k \alpha_k^{(l)} A_k^{(l)}\right)$
   (e) Upsample $L_{\text{Grad-CAM}}^{(l)}$ to input resolution $\rightarrow \mathcal{H}_c^{(l)}$
3. Return Grad-CAM heatmaps $\{\mathcal{H}_c^{(1)}, \mathcal{H}_c^{(2)}, \mathcal{H}_c^{(3)}, \mathcal{H}_c^{(4)}\}$

CSP-SegNet architecture, enabling a direct visual assessment of improving anatomical consistency and boundary smoothness. These qualitative ablation results are coherent with the results exhibited in Table 4.

The qualitative trends in Fig. 19 are highly consistent with the quantitative ablation results reported in Table 4. The backbone model (C1) yields the poorest segmentation, exhibiting anatomically inconsistent OC–OD delineations with substantial missed regions and blank segmentation masks. Introducing CSP in C2 produces a noticeable visual improvement, with better localization of OC and OD, although the boundaries remain coarse and insufficiently smooth. Configuration C3 further enhances performance on high-contrast fundus images but struggles on diffused and low-contrast samples. C4, which incorporates only IRB, generates highly irregular and unstable OC–OD contours, while C5 and C6, despite improving region coverage, suffer from erroneous outlier pixel predictions and distorted boundaries, reflecting weak structural consistency. C7 exhibits improved localization but still produces non-smooth and fragmented contours. In contrast, the full CSP-SegNet configuration (C8) delivers anatomically coherent and sharply defined OC–OD boundaries across diverse imaging conditions, closely matching the ground truth. A key observation across all configurations is the decisive role of CSP as its inclusion in every configuration produces substantially reliable and stable segmentations, whereas those without CSP fail to capture the OC–OD structure effectively. This visually corroborates the dominant contribution of CSP to the final performance gains observed in the quantitative ablation study. These qualitative and quantitative ablation results collectively confirm that the CSP, SEB, and IRB modules operate synergistically within CSP-SegNet, where their coordinated interaction yields anatomically consistent, stable, and high-fidelity OC–OD segmentation that cannot be achieved by any component in isolation.

4.7 Explainability for CSP attention using Grad-CAM

To explain the focus of our proposed CSP attention module we have generated heatmaps by exploiting Gradient-based Class Activation Mapping (Grad-CAM) [17]. This essentially improves the interpretability of our CSP attention mechanism, which is integrated within the CSP-SegNet's skip connections. The residual connections in Fig. 3 join the encoder to the decoder at four levels in the CSP-SegNet hierarchy. Consequently, it is crucial to visualize the attention maps at each level to comprehend the regions-of-focus and how it gets progressively refined. In order to interpret what OC-OD spatial features were captured that attributed for a particular segmentation result, the Grad-CAM generated heatmaps can be visualized for both OC and OD. Algorithm 2 describes an end-to-end training pipeline adopted for Grad-CAM-based Attention Visualization with CSP-SegNet.

The Grad-CAM visualizations for OC and OD are given in Figs. 20 and 21, respectively to assess the evolution of attention maps from high level to low level.

In Figs. 20 and 21 the levels correspond to each skip connection as demonstrated in Fig. 3. It is pertinent to note that Grad-CAM is applied to visualize the attention-integrated features i.e., the feature map obtained after concatenation of corresponding CSP-attention map with the upsampled decoder-level features. The schematic representation in Fig. 3 clearly demonstrates the exact decoder point from which these Grad-CAM attention maps are extracted, highlighting that the visualizations originate from the fused CSP–decoder representation rather than from a single isolated layer, thereby providing a more faithful depiction of how the network attends to OC–OD structures. From Figs. 20 and 21 we may conclude that our novel CSP attention module is capable of

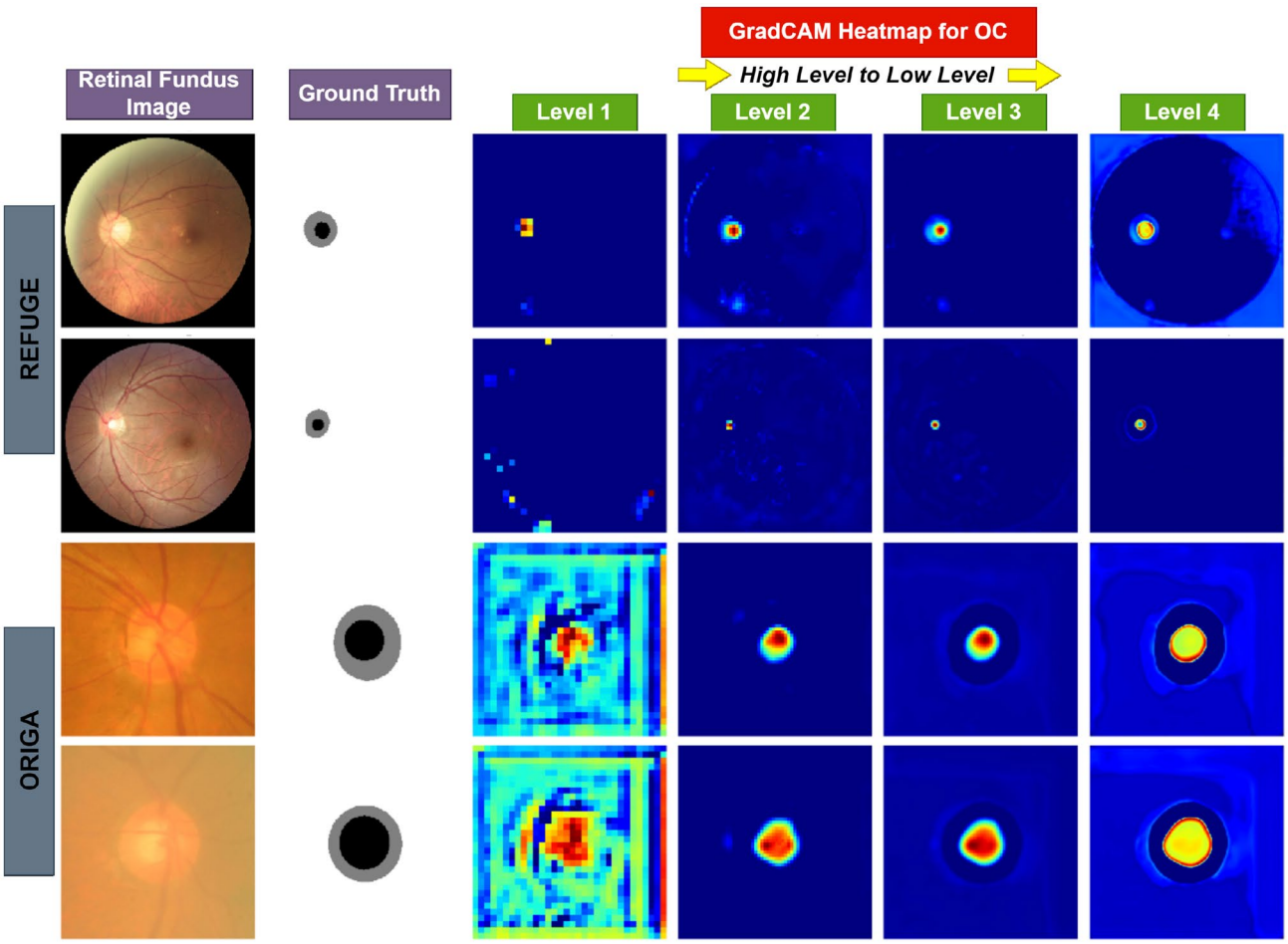


Fig. 20 Grad-CAM generated heatmap for OC

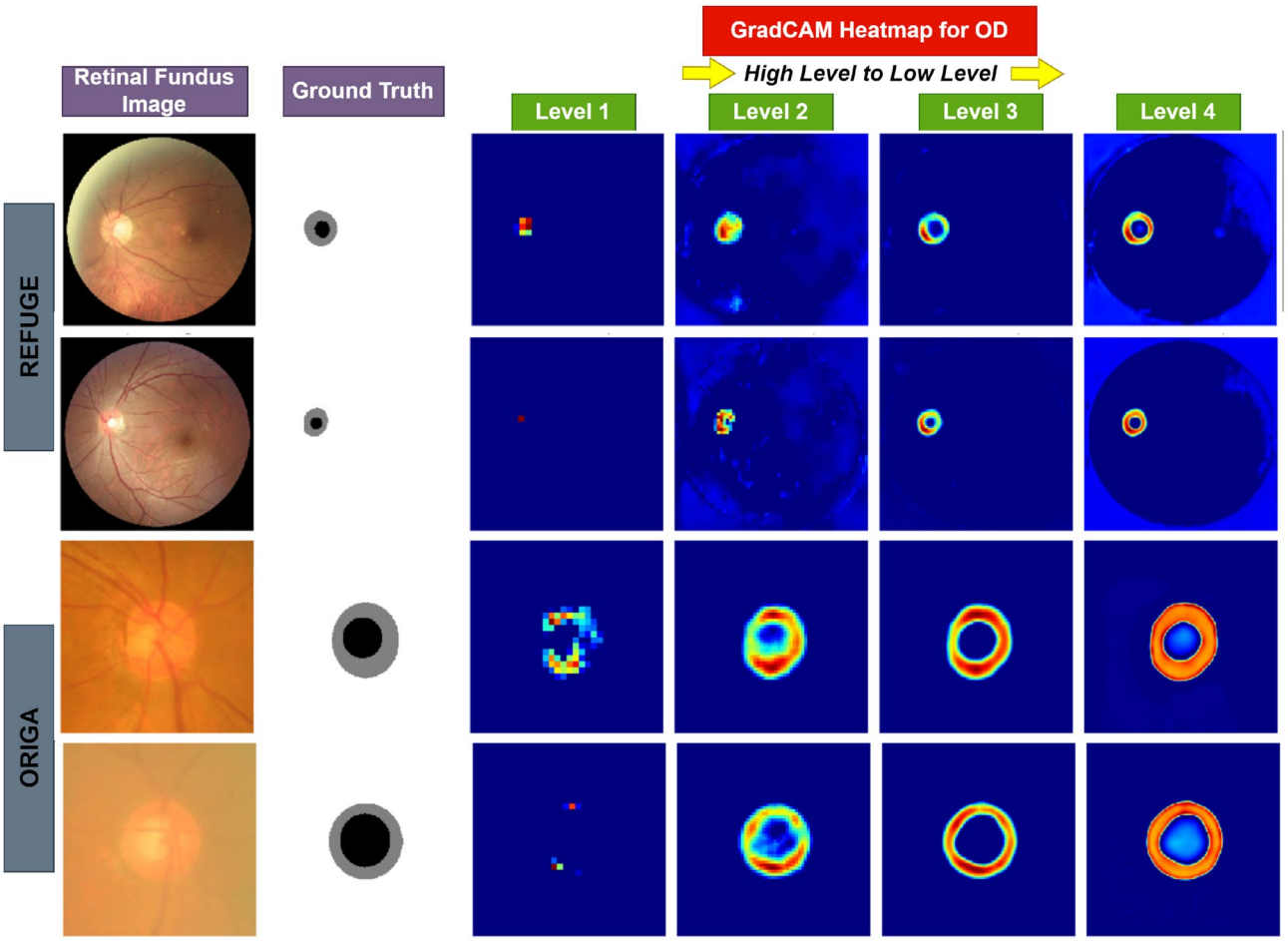


Fig. 21 Grad-CAM generated heatmap for OD

highlighting important OC-OD spatial features. Furthermore, it is observed that these attention-integrated feature maps are refined as we progress from level 1 to level 4 i.e., from high-level to low-level within the CSP-SegNet hierarchy. Hence the CSP module contributes immensely to accurate semantic segmentation of OC and OD in retinal fundus images.

4.8 Instances of failure

The datasets utilized in this study comprises heterogeneous retinal fundus images captured using different fundus imaging cameras. Hence, it is critical for any deep learning framework to precisely segment all test data points. Owing to this intricacy our proposed CSP-SegNet fails to accurately segment OC and OD in some cases as demonstrated in Fig. 22.

Our contour-based visual analysis, shows that CSP-SegNet and other advanced segmentation algorithms have failed miserably in scenarios where the OC-OD regions lack clear distinguishable boundaries. The primary causes behind such failures is the lack of brightness, poor contrast and obscure boundaries. Therefore, all deep learning frameworks including CSP-SegNet, generates undesirable segmentation masks for instances resembling indiscernible OC-OD contours especially on fundus images that are dimly lit and low contrast or have excessive highlights and indistinct details.

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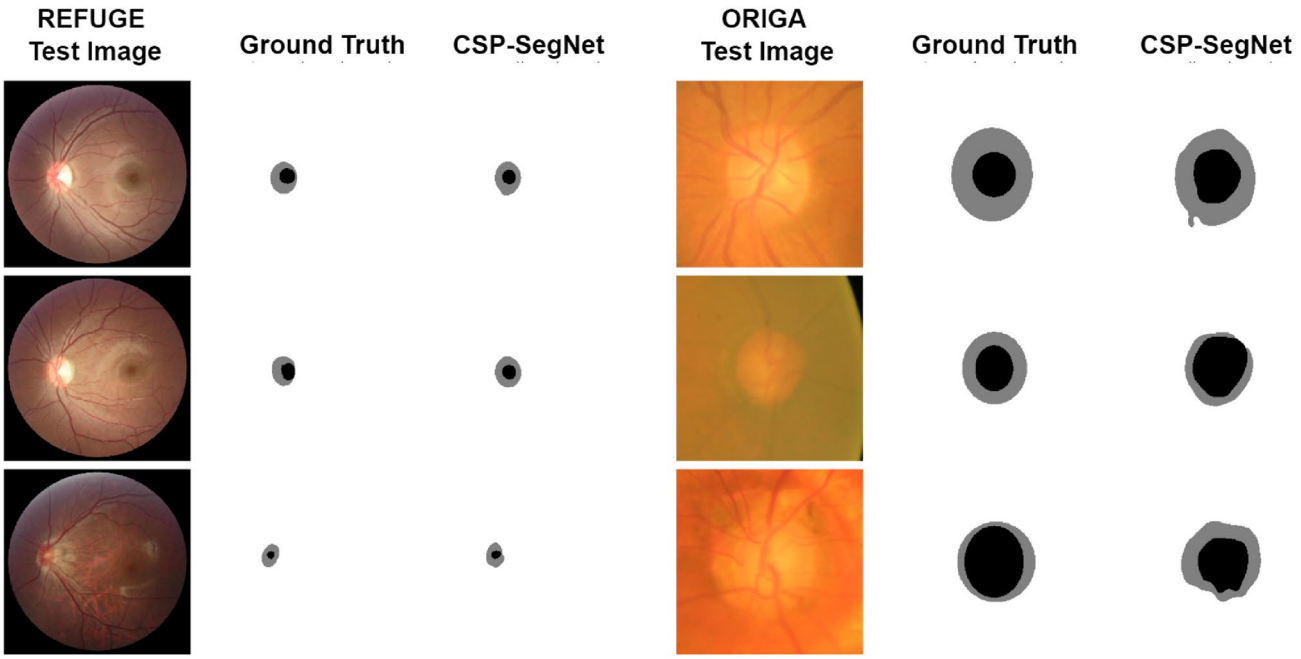


Fig. 22 Representative failure cases of CSP-SegNet highlighting limitations in OC–OD segmentation

5 Conclusion

The glaucoma disease has shattered innumerable dreams, and its threat looms large among people of different age groups. It is quintessential to improvise medical imaging applications so as to automate the detection of glaucoma during its early stages, given its subtle and almost non-existent symptoms. In line with this agenda, accurate and automatic segmentation of optic cup and optic disk in retinal fundus images can reform the current diagnostic system for early diagnosis of glaucoma through biomarker discovery. In this study, an end-to-end deep learning framework named CSP-SegNet is proposed for the joint multiclass semantic segmentation of OC and OD in retinal fundus images. CSP-SegNet is a lightweight CSP attention-integrated, novel depthwise separable convolution based encoder-decoder architecture. Our novelty lies essentially in the lightweight architecture and the proposed CSP attention mechanism and its interpretability. The utility of depthwise separable convolution ensures optimization of the computational load by substantially reducing the number of trainable parameters. CSP-SegNet comprises only 1.54M parameters and 13.3 GFLOPS, which accounts for its low computation load and complexity and hence, it is suitable for deployment into mobile-friendly medical imaging applications. The CSP attention module captures important contextual OC-OD features from channel, spatial and pixel levels of abstraction. Explaining the focus of CSP attention using Grad-CAM generated heatmaps is of utmost significance. The encoder incorporates a Squeeze and Excitation Block (SEB) followed by an Inverted Residual Block (IRB) to encode fine details from squeezed and excited features. Besides, the residual connections ensure the flow of high-level gradients and the preservation of refined features. The proposed method has achieved statistically significant segmentation results in terms of Dice Similarity Coefficient (DSC) and Intersection-over-Union (IoU) metrics on REFUGE, ORIGA and cross-dataset evaluation on Drishti-GS datasets. Therefore, the robustness and generalization ability of CSP-SegNet with respect to distribution shift is justified as it exhibits statistically significant segmentation results outperforming most state-of-the-art methods on these advanced retinal fundus imaging datasets. Moreover, the quantitative and qualitative ablation study proves that there exists a high degree of cohesiveness among the constituent modules of CSP-SegNet and therefore the CSP-SegNet architecture as a whole is collectively responsible for achieving outstanding segmentation performance. It is pertinent to note that CSP-SegNet converges significantly fast and this advantageous property acts as a differential. However,

CSP-SegNet failed in scenarios where the OC and OD boundaries were impeded by each other. That being said, our next objective is to overcome this shortcomings. The proposed method can produce accurate and reliable segmentation results and therefore has the potential to assist ophthalmologists in the diagnosis of patients.

Author contributions Ankit Das: Conceptualization, Methodology, Software, Visualization, Investigation and Writing - Original Draft. Saubhik Bandyopadhyay: Methodology, Formal Analysis, Supervision, Writing - Review & Editing. Debapriya Banik: Supervision, Formal Analysis, Validation, Writing - Review & Editing. Debotosh Bhattacharjee: Supervision, Conceptualization, Project Administration, Resources, Validation, Writing - Review & Editing.

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Data availability Data used in this research is publicly available at https://zenodo.org/records/8009107.

Code availability Code is available at https://github.com/AIAnkitDas/CSP-SegNet/tree/main.

Declarations

Conflicts of interest On behalf of all authors, the corresponding author states that there is no conflict of interest. All authors declare that they have no known conflicts of interest in terms of competing financial interests or personal relationships that could have an influence or are relevant to the work reported in this paper.

Ethical approval We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. We further confirm that the manuscript has been read and approved by all named authors. We also confirm that the order of authors listed in the manuscript has been approved by all of us. The research did not involve direct interaction with human participants or animal subjects, as the study was based solely on existing medical image data. No ethical conflicts or issues were encountered.

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